

The Neutrino-Mass Low-Energy Exact-Character Endpoint

Mass-Role Fixation, Exact-Dirac Branch Exclusion, and Majorana-Compatible Selection by Exhaustion

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Abstract

This paper treats the origin and exact character of neutrino mass as a constraint-exhaustion problem. The Admissibility And Standing Constraint framework is used as a mathematical constraint formalism, not as an additional neutrino model or an interpretive overlay. Its proof tools are exhaustion, impossibility, deletion, quotient collapse, and no-promotion. The upstream kernel fixes the minimal conditions under which non-degenerate construction and standing-bearing sameness have meaning. Realized-persistence, field-carrier, and mass-role results then determine what a physically meaningful neutrino-mass claim must occupy before a Dirac, Majorana, Weinberg, singlet, or ultraviolet representation can compete.

The first positive result is a spectral-mass-role bridge. In the declared coherent PMNS mass-driven scope, a nontrivial oscillation phase cannot be discharged by a scalar label, matrix notation, measurement record, or operator availability alone. It certifies the neutrino mass-role target, the quotient-stable deformation/transport-response profile $\mu_D(\Pi_\nu)$ of the neutrino persistence class. Candidate operators are therefore audited as possible occupants of an already fixed role rather than as lower generators of that role.

Theorem 4.5 derives the current corpus-snapshot low-energy active-neutrino domain $\mathfrak{D}_{\nu,0}^{\text{LE}}$ from the admitted witness family, certified carrier inventory, and low-energy quotient before exact-character branches are evaluated. The minimal renormalizable Standard-Model envelope is deleted because it contains no active-neutrino mass-role occupant. The Standard-Model-field envelope through dimension five contains the unique Weinberg class. A low-energy exact-Dirac alternative would require both a certified Dirac-completing field-carrier package and a standing-bearing protector excluding every admitted $\Delta L = 2$ route in the declared envelope. Exhaustion over titles, objects, claims, witnesses, retrieval terms, and dependency records finds no such certified carrier-protector package in the current corpus snapshot. A written N_R , a chosen vanishing Majorana coefficient, an uncertified global label, or a finite null search does not satisfy that burden.

The resulting theorem is a certified low-energy exact-character collapse relative to $\mathfrak{D}_{\nu,0}^{\text{LE}}$. Exact Dirac preservation is not a certified same-target survivor of the current corpus-snapshot target $\mathfrak{D}_{\nu,0}^{\text{LE}}$. Richer-scope routes without a surviving witness in the declared low-energy witness family quotient-collapse to their low-energy residue; an exact-Dirac rescue is handled by the update protocol $\mathfrak{D}_{\nu,+}^{\text{LE}}$ and requires its own carrier, protector, matching, and witness certificates. Here $\text{MajoranaCompatible}_{\text{LE}}$ denotes the low-energy exact-character quotient in which the occupied active-neutrino mass residue fails exact preservation of Λ_ν ; it does not denote a unique ultraviolet mechanism or a pure-Majorana ultraviolet representative. The selected low-energy exact-character endpoint in $\mathfrak{D}_{\nu,0}^{\text{LE}}$ is $\text{MajoranaCompatible}_{\text{LE}}$.

The theorem is AASC-internal by construction. A same-target counterexample must break a load-bearing dependency, derive a kernel role from below without presupposing governance-

equivalent work, or preserve the same target, mass role, witness family, envelope discipline, and exact-character selector while producing a contrary certified endpoint.

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Executive result and claim classes

The dependency and proof spine is

$$\begin{aligned} \text{TargetAdeq}_D &\implies K_D \implies \text{RealizedPersistence} \implies \Phi_D \implies \mu_D(\Pi_\nu), \\ W_{\text{osc}}^{m,\text{PMNS}} &\implies \text{RoleTarget}_{\mathfrak{D}_{\nu,0}^{\text{LE}}}^{\text{cert}}(\mu_D(\Pi_\nu)) = 1 \implies \text{EnvelopeExhaustion} \implies \text{ExactCharacterCollapse}. \end{aligned} \quad (1)$$

The intermediate arrows are role dependencies, not claims that this paper rederives the complete realized-physics arc. The kernel theorem is fixed by the canonical corpus lineage and is independently restated in self-contained form in the foundational burden source; R1, persistence-regime realization, R6, and R9 supply the downstream realized-persistence, carrier, and mass-role layers [14, 15, 16, 17, 18, 19].

The principal endpoint is

$$\begin{aligned} W_{\text{osc}}^{m,\text{PMNS}} &\implies \text{RoleTarget}_{\mathfrak{D}_{\nu,0}^{\text{LE}}}^{\text{cert}}(\mu_D(\Pi_\nu)) = 1, \\ \text{ExactDirac}_{\mathfrak{D}_{\nu,0}^{\text{LE}}} &\implies \text{CarrierPkg}_{\nu_R}^{\text{cert}} \wedge \text{ProtectorPkg}_{\Delta L}^{\text{cert}}, \\ \neg(\text{CarrierPkg}_{\nu_R}^{\text{cert}} \wedge \text{ProtectorPkg}_{\Delta L}^{\text{cert}}) &\implies \text{Surv}_{\mathfrak{D}_{\nu,0}^{\text{LE}}}^{\text{cert}}(\text{ExactDirac}) = 0, \\ \text{LowEnergyClosure} &\implies \text{Sel}_{\mathfrak{D}_{\nu,0}^{\text{LE}}}(\text{MajoranaCompatible}_{\text{LE}}) = 1. \end{aligned} \quad (2)$$

Here **LowEnergyClosure** abbreviates the explicit assumptions of Theorem 12.5: mass-role target certification, deletion of $\mathcal{E}_0$, uniqueness of the Weinberg low-energy active-mass class, low-energy quotient collapse of unwitnessed richer-scope routes, and the corpus-audited absence of a certified Dirac carrier–protector package.

Definition 0.1 (Majorana-compatible low-energy exact character). *MajoranaCompatible_{LE} is the low-energy exact-character quotient in which the occupied active-neutrino mass-role residue fails exact preservation of the nontrivial quotient-stable additive neutrino/antineutrino persistence relation Λ_ν . The definition makes no claim of a unique ultraviolet mechanism, unique heavy carrier, or pure-Majorana ultraviolet representative. In particular,*

$$\text{MajoranaCompatible}_{\text{LE}} \not\equiv \text{PureMajorana}_{\text{UV}}. \quad (3)$$

Table 1: Claim classes used in this paper.

Class	Meaning
Imported role theorem	An AASC theorem, self-contained proof, or corpus-certified exact-object dependency fixing an upstream role without importing a neutrino mechanism.
Standard field-theory input	A conventional operator, gauge, Lorentz, or neutral-spinor fact used to define the finite comparison arena.
Constraint-exhaustion theorem	An impossibility, deletion, quotient, burden, or endpoint theorem obtained by exhausting the declared routes.
Corpus-certified endpoint	Selection relative to the active source pool, declared field-carrier inventory, source lock, and low-energy witness family.

Class	Meaning
Scope boundary	A stronger claim intentionally excluded because it changes the target or lacks a standing-bearing certificate.

Source note 0.2 (Source and corpus-audit rule). The project archive contains the oscillation, NNR, CNF, Standard-Model, field-carrier, persistence-regime, mass-role, and foundational-burden sources used for proof-sensitive reuse. The canonical kernel lineage is represented by exact accepted-object and version/dependency records. For direct inspection, the project archive includes a provenance-marked, page-exact extract of the complete self-contained kernel proof together with the full foundational-burden source from which it is taken. The exact-Dirac absence result is supported by a reproducible carrier/protector query audit over the integrated corpus, not by an unrecorded impression that no protector exists.

1 Introduction

1.1 Mathematical-status lock and proof-class boundary

AASC is used here as a mathematical constraint formalism for fixed-domain role occupancy, selector standing, and branch elimination. The proof class is a kernel-forced low-energy exact-character exhaustion theorem. It is not a phenomenological fit, a preference ranking among neutrino models, or a request that standard field theory adopt unfamiliar terminology. Standard operator and oscillation facts define the comparison arena; AASC determines which promotions, occupants, and branch claims retain standing in that arena. The proof-class and same-target closure architecture follows the kernel-first discipline developed in the foundational burden analysis [15].

Proposition 1.1 (NMC-P11: mathematical-status and same-target admissibility lock). *AASC dependence does not supply a governance-non-equivalent same-target alternative to an AASC theorem. Any contrary same-target construction must invalidate a definition, theorem link, dependency transport, target specialization, source certification, or final inference; alternatively, it must derive a load-bearing kernel role from below or realize a faithful same-regime counterexample.*

Proof. Every theorem is evaluated inside a declared proof setting. Dependence fails only when a dependency is false, misapplied, circularly transported, or insufficient for the conclusion. A change of vocabulary, proof route, or model scope does not alter the theorem chain unless it changes a load-bearing invariant. The two substantive foundational alternatives are lower derivation and same-regime dispensability; these are formalized in Section 16. $\square$

1.2 The constraint target

Neutrino oscillation requires nontrivial propagation-spectrum structure, yet the minimal renormalizable Standard Model contains no active-neutrino mass operator. Standard constructions add right-handed singlets, admit the dimension-five Weinberg operator, introduce seesaw sectors, or build radiative and other richer-scope mechanisms. Those constructions establish writable possibilities. They do not by themselves establish which possibility has standing in the realized low-energy target, which field carrier is certified, which exact protector exists, or which ultraviolet distinctions survive low-energy quotienting.

The target of this paper is therefore not another mass model. It is the standing of the promotion from a field-theoretic possibility to a selected physical claim. The operative question is:

Given the corpus-certified low-energy active-neutrino mass-role target, can exact Dirac preservation remain a certified survivor without a certified Dirac-completing carrier and an all-operator standing protector?

Definition 0.1 fixes the exact low-energy endpoint class before the operator and carrier branches are evaluated.

1.3 AASC as a constraint formalism

AASC is a constraint formalism. Its native proof modes are exhaustion, impossibility, deletion, quotient collapse, and no-promotion. It does not add particles or interactions. It removes alternatives that cannot preserve the declared target without hidden load, unlicensed repair, selector import, or scope change. A negative conclusion is not an incomplete positive model. It is the theorem output when a proposed route cannot perform the role it claims while preserving the comparison object.

Impossibility is the native output of the formalism when a proposed route cannot preserve the declared target. Once the exact-character selector and its burden are fixed, bivalence requires testing whether both branches are realizable in the evaluated target. The exact-Dirac branch survives inside the evaluated target only if its carrier and protector obligations are discharged. If they are not, it is excluded from the certified same-target survivor set of that target rather than retained as an ungrounded equal.

1.4 Dependency direction: kernel, realization, carrier, and mass role

The dependency direction is upstream from the AASC kernel to physical claim status. The direct kernel result identifies admissibility, standing, reference, and irreversibility as the minimal mutually closed conditions for non-degenerate construction and same-target reuse. R1 prevents realized persistence from collapsing into null reconstruction. Persistence-regime realization blocks trace-only, abstract-only, repairable, and skin-only participation. R6 fixes the distributed carrier role required for standing-bearing transport. R9 fixes mass as the operative deformation-response profile represented by later scalar masses, mass matrices, Yukawa entries, and effective operators [14, 16, 17, 18, 19].

Accordingly, a model cannot be used as a lower generator of the target, field carrier, mass role, sameness relation, or selector by which it wins. A written N_R symbol is not yet a certified carrier, and an imposed symmetry label is not yet a standing protector.

1.5 Dependency-inversion exclusion

The relevant dependency is not

$$\text{AASC framework} \implies \text{neutrino-character restriction.}$$

It is

$$\begin{aligned} &\text{determinate same-regime neutrino} \\ &\text{mass-character evaluation} \implies K_D \implies \text{AASC consequence layer.} \end{aligned}$$

A determinate comparison already fixes a target, permits evaluable branch claims, distinguishes continuation from repair, and preserves the comparison under lawful redescription. These are the kernel-neutral adequacy roles later named reference, standing, irreversibility, and admissibility.

Theorem 1.2 (NMC-T11: dependency-inversion exclusion). *A proposed kernel-independent construction that retains the same determinate neutrino mass-character target reverses the dependency order. It can survive only by showing that one of the target-adequacy roles is unnecessary, deriving the corresponding kernel work from a strictly lower non-circular basis, or realizing a faithful same-regime counterexample that preserves the same target and comparison class while omitting that work.*

Proof. Target determinacy, branch evaluability, act-time finality, and same-regime fidelity are required before a Dirac, Majorana-compatible, or richer-scope claim can be evaluated as a claim about the same object. Granting those roles grants the governance work of the kernel under whatever terminology is used. Denying one of them loses the corresponding fixed-target good. Reintroducing the lost good through an auxiliary selector, hidden carrier, repair channel, or target translation imports governance-equivalent work or changes scope. Hence the kernel is upstream of the neutrino comparison rather than an optional downstream classifier. $\square$

1.6 Targets that must remain separate

The phrase “origin of neutrino mass” merges several targets:

1. mass-driven spectral nontriviality;
2. certification of the mass-role target $\mu_D(\Pi_\nu)$;
3. selection of the low-energy operator occupant;
4. exact Dirac versus Majorana-compatible character;
5. mixed, rank, or pseudo-Dirac block refinement;
6. field-envelope realization; and
7. ultraviolet origin and scale.

The present paper certifies the first two, exhausts the low-energy operator and exact-character branches, and selects a Majorana-compatible low-energy endpoint relative to $\mathfrak{D}_{\nu,0}^{\text{LE}}$. It does not identify a unique ultraviolet route or numerical mass spectrum.

1.7 Standard inputs and AASC outputs

The field-theory ingredients are deliberately standard. Novelty is not claimed for the Weinberg operator, the absence of an active mass term in the minimal renormalizable field load, or the simultaneous availability of Dirac and Majorana singlet terms. The theorem object is the status of the promotion made from those facts.

Table 2: Standard comparison inputs versus AASC constraint outputs.

Standard comparison input	AASC constraint output
Mass-driven oscillation has nonzero split-support.	The neutrino mass-role target is certified; no specific operator or character follows from oscillation alone.
The minimal renormalizable Standard Model has no active-neutrino mass operator.	$\mathcal{E}_0$ is deleted as a complete realized envelope once the mass-role witness is admitted.
The Weinberg class is unique through dimension five with Standard-Model light fields.	It is the unique certified low-energy active-neutrino mass-role occupant after the carrier inventory and quotient target are fixed.
A gauge-singlet N_R admits a Dirac Yukawa term.	Exact Dirac requires a certified N_R -type carrier and a standing protector over every admitted $\Delta L = 2$ route.
Standard oscillation probabilities are insensitive to Majorana column phases.	Oscillation is not an exact-character selector.
Different ultraviolet mechanisms can match to the same low-energy residue.	They are one low-energy character class unless a witness in $\mathcal{W}_{\text{low}}$ survives.
A finite $\Delta L = 2$ search can return null.	Non-observation cannot be promoted to an exact all-operator protection theorem.
The corpus contains no certified Dirac carrier-protector package.	Exact Dirac is excluded from the certified same-target survivor set of $\mathfrak{D}_{\nu,0}^{\text{LE}}$; MajoranaCompatible _{LE} is selected in that target.

1.8 Main results

The paper proves five levels of result.

- (i) **Spectral-mass-role bridge.** $W_{\text{osc}}^{m,\text{PMNS}}$ certifies $\mu_D(\Pi_\nu)$ as the relevant target, rather than a primitive scalar or matrix label.
- (ii) **Operator occupant exhaustion.** $\mathcal{E}_0$ has no occupant; $\mathcal{E}_5$ has the unique Weinberg class; $\mathcal{E}_{\nu_R}$ requires new carrier load and admits Dirac and Majorana-compatible blocks.
- (iii) **Exact-character normal form.** For a neutral fermion mass response, the exact axis is preservation or failure of a quotient-stable additive particle/antiparticle persistence distinction; mixed and pseudo-Dirac structures refine the violated branch rather than supply a third exact value.
- (iv) **Dirac burden and certified absence.** Exact Dirac requires both $\text{CarrierPkg}_{\mathcal{S}_{\nu_R}}^{\text{cert}}$ and $\text{ProtectorPkg}_{\Delta L}^{\text{cert}}$; the corpus-wide audit certifies no such package in $\mathfrak{D}_{\nu,0}^{\text{LE}}$.
- (v) **Low-energy collapse.** After deleting $\mathcal{E}_0$, quotient-collapsing unwitnessed richer-scope routes, and deleting exact Dirac, the selected exact-character endpoint is **MajoranaCompatible**_{LE}.

1.9 Nonclaims and scope boundaries

Scope statement 1.3 (Numerical firewall). This paper does not derive the absolute masses, mass sum, PMNS coordinates, a seesaw scale, a Weinberg coefficient, or a neutrinoless-double-beta decay rate.

Scope statement 1.4 (Ultraviolet firewall). The selected low-energy exact-character endpoint in $\mathfrak{D}_{\nu,0}^{\text{LE}}$ is `MajoranaCompatibleLE`. It is not a proof of a unique ultraviolet completion, a unique heavy carrier, or a pure-Majorana ultraviolet mass matrix.

Scope statement 1.5 (Target-scope firewall). The global claim is global relative to the current corpus-snapshot target $\mathfrak{D}_{\nu,0}^{\text{LE}}$, not every logically writable extension. A richer-scope exact-Dirac rescue is handled by the update protocol $\mathfrak{D}_{\nu,+}^{\text{LE}}$ and must supply its own certified carrier, protector, matching, and surviving witness package.

2 Constraint kernel, target lock, source lock, and audit method

2.1 Kernel necessity and dependency direction

The construction-level dependency is

$$\text{TargetAdeq}_D \implies K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D). \quad (4)$$

Reference fixes what is being constructed; standing makes successful content reusable; admissibility separates licensed continuation from failure; irreversibility prevents a failed act from being repaired as the same act. The canonical kernel lineage records mutual closure, minimality, same-target invariant discipline, and exhaustion of non-kernel alternatives. Exact accepted-object and version/dependency records are included in the project archive. The complete self-contained proof actually used for inspection is separately bundled as a provenance-marked extract of Section 3 of the foundational-burden source, and the full source containing that proof is also included [14, 15].

Proposition 2.1 (NMC-P0: dependency-direction lock). *Every neutrino-mass selection claim in this paper is downstream of kernel adequacy, realized-persistence target fixation, field-carrier mediation, and mass-role target fixation. No Dirac, Majorana, Weinberg, singlet, or ultraviolet model may define the sameness relation or selector by which it is compared.*

Proof. Removing reference destroys target identity; removing standing destroys reuse; removing admissibility destroys licensed comparison; removing irreversibility permits same-act repair. R1, persistence-regime realization, R6, and R9 then show respectively that the physical target cannot be null, realized role participation cannot be trace-only, field content must mediate transport, and mass parameters must represent the mass-role profile. A proposed neutrino model therefore enters only after these roles are fixed. $\square$

2.2 The corpus-certified low-energy target

Definition 2.2 (Corpus-certified low-energy active-neutrino target family). *The symbol $\mathfrak{D}_{\nu}^{\text{LE}}$ denotes the low-energy active-neutrino exact-character target schema. Its current source-state instance is $\mathfrak{D}_{\nu,0}^{\text{LE}}$, the fixed evaluated target consisting of:*

1. *the active-neutrino boundary and propagation witness family fixed by CS6C and NNR;*

2. the corpus-certified Standard-Model field-carrier and bridge inventory present in the current corpus snapshot;
3. the mass-role target $\mu_D(\Pi_\nu)$;
4. the finite envelopes $\mathcal{E}_0$ and $\mathcal{E}_5$, with no additional envelope admitted unless its new carrier load is certified in the current source state;
5. the declared low-energy quotient π_{low} and witness family $\mathcal{W}_{\text{low}}$; and
6. fail-closed treatment of hidden carriers, uncertified protectors, coefficient stipulations, and scope-changing rescue routes.

The update protocol $\mathfrak{D}_{\nu,+}^{\text{LE}}$ is the rule for a later source state in which an additional field carrier, protector, matching map, or surviving low-energy witness is certified. Such an update triggers a new target-fixation and carrier/protector audit. It does not refute the theorem proved here for $\mathfrak{D}_{\nu,0}^{\text{LE}}$.

Definition 2.3 (Envelope selection). $\text{EnvSel}_\nu(\mathcal{E}) = 1$ means that $\mathcal{E}$ is certified as the realized comparison arena for the specified target instance. For $\mathfrak{D}_{\nu,0}^{\text{LE}}$, envelope selection may be certified only by impossibility deletion of rivals, a completeness certificate inside the current certified carrier inventory, a standing-bearing new-carrier witness already present in the source state, a surviving low-energy witness, or a protector certificate. Preference, naturalness, and notation have no envelope-selector authority. A future certification belongs to $\mathfrak{D}_{\nu,+}^{\text{LE}}$ until the target is re-audited.

2.3 Formal status calculus

Definition 2.4 (Role target, availability, occupancy, survival, and selection). For a fixed evaluated target X and a candidate class C , define:

$\text{RoleTarget}_X^{\text{cert}}(R) = 1$	The theorem/witness stack certifies R as the physical role to be occupied, without yet identifying an occupant class.
$\text{Avail}_X(C) = 1$	C satisfies the declared field, gauge, Lorentz, locality, dimension, and standing constraints.
$\text{Occ}_X^{\text{real}}(C) = 1$	The realized physical regime actually contains a nonzero role occupant in C .
$\text{Occ}_X^{\text{cert}}(C) = 1$	The admitted theorem/witness stack certifies that C occupies the role.
$\text{Surv}_X(C) = 1$	C is available, not eliminated, and not quotient-collapsed. Survival alone does not imply occupancy.
$\text{OccSurv}_X(C) = 1$	C is both a survivor and certified occupied.
$\text{Sel}_X(C) = 1$	C is the unique certified occupied survivor quotient at the declared target.
$\text{Rich}_X(C) = 1$	C changes a standing-bearing component of the target or its envelope.

The physical evaluated status remains bivalent once the target is fully evaluated. “Not certified by this source stack” is an audit status, not a third physical status.

Criterion 2.5 (NMC-K1: standing-preserving selection criterion). *For a fixed target X and equivalence relation $\sim_X$,*

$$\text{Sel}_X(C) = 1 \iff \text{Avail}_X(C) = 1 \wedge \text{Occ}_X^{\text{cert}}(C) = 1 \wedge \forall C' \not\sim_X C, \text{OccSurv}_X(C') = 0. \quad (5)$$

Proof. Selection is uniqueness over certified occupied survivor quotients, not over writable classes. Availability, certified occupancy, and closure of every inequivalent occupied rival are jointly necessary and sufficient relative to the declared target. $\square$

Definition 2.6 (Additive neutrino persistence relation). *Let A_ν be an admitted additive charge/persistence-value object and let*

$$\Lambda_\nu : \Pi_\nu \longrightarrow A_\nu \quad (6)$$

be quotient-stable under gauge, basis, boundary, and transport redescription. The verdict $\text{Stand}_{\Delta L, \mathcal{E}}(\mu_D(\Pi_\nu)) = 1$ means every admitted occupied mass-role operation in $\mathcal{E}$ preserves Λ_ν ; the value is 0 when an occupied $\Delta L = 2$ route violates it. A conventional symbol L is not the relation and has no independent selector authority. Λ_ν is not assumed as a primitive conserved global charge. It is the minimal diagnostic relation whose preservation would be required for exact Dirac standing; if no admitted field-carrier/protector package realizes it as standing-bearing, it functions as the failed exact-Dirac burden rather than as an independently occupied symmetry.

2.4 Source and corpus lock

Table 3: Focused source lock and permitted reuse.

Source	Imported role	Reuse boundary
Kernel lineage and bundled proof extract	Target adequacy forces the kernel roles; mutual closure, minimality, non-derivability from below, quotient identity, and no silent re-description	Exact corpus rows and version meta-data are included. A provenance-marked extract reproduces the complete self-contained proof used here from the bundled foundational-burden source; no uninspected kernel step is imported.
Foundational burden source	Mathematical-status lock, dependency-inversion exclusion, kernel-denial costs, weakening resistance, and faithful same-regime counterexample conditions	Only the general kernel and same-target closure architecture is reused; the source’s SAT-specific endpoint conclusion is not imported.
R1	Realized persistence cannot be reconstruction-null	Does not derive fields, particles, mass, or a neutrino mechanism.
Persistence-regime realization	Irreversible standing-bearing continuation induces consequence-retaining persistence-regime equivalence	Does not select a concrete representative or neutrino operator.

Source	Imported role	Reuse boundary
R6	Localized persistence under bounded transport requires an operative carrier assignment Φ_D	A written field symbol is not a certified carrier.
R9	Identity-preserving deformation induces $\mu_D(\Pi)$; scalar masses and matrices are later representations	Does not derive neutrino values or character.
CS6C and NNR1	Mass-driven boundary kernel and symbolic splitting witness	No absolute scale, sterile exclusion, or Dirac/Majorana verdict.
NNR3 and NNR0.8	Difference-space versus shape and anchor firewalls	No numerical shape or character selection.
CNF6, CNF7, MGN5	Matrix support, quotient spectrum, phase, and basis discipline	No character occupancy from diagonalization alone.
MGN3 and MGN4	Bridge mediation and Yukawa slot availability	No nonzero coupling, new carrier, or protector is certified by slot availability.
Charge, SM, generation	CS3, Gauge/charge/anomaly skeleton, finite alternatives, and hidden-selector discipline	The two neutrino-singlet remarks describe options; they do not certify an occupied N_R carrier or exact protector.
Weinberg standard EFT	Unique SM-field dimension-five active-neutrino mass class	Does not identify a unique ultraviolet origin.
Corpus carrier and protector audit	Full active-pool search over title, object, claim, retrieval, target-role, role-signature, and witness fields	Finds singlet-option remarks but no certified Dirac carrier plus all-operator protector theorem.

2.5 Audit protocol

Definition 2.7 (Joint carrier–protector audit acceptance criterion). *A corpus object counts as a certified exact-Dirac rescue only if it jointly certifies all three of the following in one dependency-compatible route:*

1. *a realized N_R or equivalent Dirac-completing field carrier with field-role standing in the declared target;*
2. *certified occupation of the corresponding Dirac mass response; and*
3. *a standing-bearing protector covering every admitted $\Delta L = 2$ route in the declared envelope, or the admitted operator tower when that envelope is unbounded.*

Availability remarks, anomaly inertness, sterile options, displayed coefficient zeros, naturalness arguments, and Dirac/Majorana alternative lists do not satisfy this joint criterion without the carrier, occupancy, and protector certificates.

The reproducible audit searched paper titles, object names, claim and witness rows, retrieval terms, target-role and role-signature fields, and dependency edges for right-handed-neutrino, sterile, singlet, Dirac, lepton-number, $B-L$, protector, $\Delta L = 2$, Majorana, and Weinberg routes. A row was counted as an exact-Dirac rescue only when the carrier, occupied Dirac response, and all-admitted- $\Delta L = 2$

protector certificates occurred in one dependency-compatible route. The audit found availability and option-level remarks but no object satisfying the joint criterion.

Every theorem and endpoint is audited through these questions:

1. Which kernel and realized-role condition makes the asserted sameness meaningful?
2. Is the statement fixing a role target or certifying a class occupant?
3. Which field carriers are certified rather than merely writable?
4. Which envelope and low-energy witness family define the target?
5. Which alternatives are deleted, quotient-collapsed, deferred, or moved to richer scope?
6. Does a purported protector cover every admitted $\Delta L = 2$ route?
7. Is an absence claim reproduced by the corpus audit rather than inferred informally?
8. Which stronger ultraviolet or numerical claim remains outside the endpoint?

The answers are mirrored in machine-readable ledgers and query extracts included in the project archive.

3 Target formation from mass-driven neutrino oscillation

3.1 Boundary kernel and invariant witness

Let $|\nu_\alpha\rangle$ denote a flavor-labeled boundary state at production and $|\nu_i\rangle$ a propagation eigenstate in the declared coherent finite PMNS scope. Write

$$|\nu_\alpha\rangle = \sum_i U_{\alpha i}^* |\nu_i\rangle, \quad K_{\beta\alpha}(L, E) = \sum_i U_{\beta i} U_{\alpha i}^* \exp\left(-i \frac{m_i^2 L}{2E}\right), \quad (7)$$

up to the ordinary relativistic approximation and an overall common phase. CS6C classifies $K_{\beta\alpha}$ as the same-scope boundary-to-boundary reconstruction object and the products

$$U_{\alpha i} U_{\beta i}^* U_{\alpha j}^* U_{\beta j} \quad (8)$$

as rephasing-invariant witness content [1]. NNR1 admits the mass-squared differences only as correctly typed witness roles, not as numerical inputs [2].

Definition 3.1 (Mass-driven oscillation witness). $W_{\text{osc}}^{m, \text{PMNS}}$ is the standard coherent PMNS mass-driven witness that the declared propagation kernel contains at least one physically nontrivial phase difference

$$\Delta m_{ij}^2 = m_i^2 - m_j^2 \neq 0 \quad (9)$$

for a pair with nonvanishing boundary mixing support. The label (m, PMNS) fixes the scope: the witness is the standard coherent PMNS mass-driven readout. Non-mass-driven flavor-conversion proposals are different targets unless they reproduce the same mass-role witness family under an explicit bridge.

3.2 Majorana phases cancel from the oscillation kernel

A Majorana mixing matrix may be written as $U = VP_M$ with diagonal column-phase matrix P_M . Then $U_{\beta i}U_{\alpha i}^* = V_{\beta i}V_{\alpha i}^*$, so Majorana column phases cancel from the standard source–propagation–detection amplitude.

Lemma 3.2 (NMC-L1: oscillation non-selection of mass character). *Within the declared coherent PMNS mass-driven scope, $W_{\text{osc}}^{m,\text{PMNS}}$ distinguishes nontrivial spectral differences but does not distinguish exact Dirac from Majorana-type light-neutrino character.*

Proof. The boundary kernel depends on mass-squared differences and products $U_{\beta i}U_{\alpha i}^*$. Rephasing of a propagation eigenstate cancels within each product, and the additional Majorana column phases are absent from the transition probability. Dirac and Majorana theories with the same m_i^2 , quotient-stable mixing products, and standard boundary interactions therefore yield the same kernel. The witness selects spectral nontriviality, not the standing of the neutrino/antineutrino distinction. $\square$

Corollary 3.3 (NMC-C1: absolute-offset deferral). *The transformation $m_i^2 \mapsto m_i^2 + c$ multiplies the kernel by a common phase and leaves transition probabilities unchanged. Oscillation does not determine the common squared-mass offset.*

4 Mass-role target fixation

4.1 The upstream physical-role stack

R1 proves that realized persistence cannot be reconstruction-null relative to its own continuation-sensitive target. Persistence-regime realization shows that irreversible standing-bearing participation cannot remain trace-only, abstract-only, or skin-level. R6 derives field structure as an operative carrier assignment. R9 derives the mass role of a particle persistence class Π as

$$\mu_D(\Pi) : \Delta_\Pi \rightarrow C_D, \quad \iota_D(\Pi) = \mu_D(\Pi)|_{\Delta_\Pi^{\text{tr}}}. \quad (10)$$

Mass parameters, Yukawa entries, running masses, and mass matrices are physically mass-relevant only insofar as they represent this role in their proper envelope [16, 17, 18, 19].

Definition 4.1 (Neutrino mass-role profile). *Let Π_ν denote the neutrino persistence class in the declared field-carrier regime. Its mass-role target is*

$$\mu_D(\Pi_\nu) : \Delta_{\Pi_\nu} \rightarrow C_D. \quad (11)$$

A neutrino mass mechanism is an admissible candidate only if it realizes or represents the relevant standing-preserving deformation/transport response of Π_ν .

Theorem 4.2 (NMC-T1: spectral–mass-role bridge and target fixation). *Within the declared standard coherent PMNS mass-driven target,*

$$W_{\text{osc}}^{m,\text{PMNS}} \implies \text{RoleTarget}_{\mathfrak{Z}_{\nu,0}^{\text{LE}}}^{\text{cert}}(\mu_D(\Pi_\nu)) = 1. \quad (12)$$

The witness certifies the mass role as the target that must be occupied; it does not select a field-theoretic occupant, exact character, absolute scale, or ultraviolet origin.

Proof. $W_{\text{osc}}^{m,\text{PMNS}}$ supplies a transport-facing nontrivial mass-squared phase difference with boundary mixing support. In R9, the transport-facing inertia role is the restriction of the broader mass-role profile. Therefore the oscillation distinction is a specialization of the deformation/transport response of the neutrino persistence class, not a primitive scalar label. R1 and persistence-regime realization block replacement of that distinction by trace or skin. R6 requires operative field-carrier mediation. R9 exhausts scalar assignment, substance amount, energy ledger, gauge artifact, hidden reservoir, measurement record, premature Standard-Model parameter, and cross-domain import. The licensed conclusion is role-target certification. Lemma 3.2 blocks promotion to a particular character occupant. $\square$

Corollary 4.3 (NMC-C2: target fixation is not operator occupancy).

$$\text{RoleTarget}_{\mathfrak{D}_{\nu,0}^{\text{LE}}}^{\text{cert}}(\mu_D(\Pi_\nu)) = 1 \not\Rightarrow \text{Occ}_{\mathcal{E}}^{\text{cert}}(C) = 1 \quad (13)$$

for any specific Dirac, Weinberg, singlet-Majorana, mixed, or richer-scope class without an envelope and occupant theorem.

Proposition 4.4 (NMC-P9: primitive mass-source collapse). *A scalar neutrino mass, mass matrix, Yukawa entry, or experimental estimate introduced before role-target fixation cannot source the role. If it tracks $\mu_D(\Pi_\nu)$, it is a representation of the derived role; if it does not, it has no selector authority.*

Proof. This is the neutrino specialization of the R9 route-exhaustion result. The physical content of the proposed object is exactly its effect on the standing-preserving response profile. No independent primitive mass essence remains after quotienting role-equivalent descriptions. $\square$

Theorem 4.5 (NMC-T16: low-energy target fixation by witness family). *Let the admitted readout be the active-neutrino boundary and propagation witness family, let the certified light-field inventory be the Standard-Model carrier and bridge inventory, and let richer descriptions be identified by the declared low-energy quotient whenever no witness in $\mathcal{W}_{\text{low}}$ survives. Then*

$$\begin{aligned} W_{\text{osc}}^{m,\text{PMNS}} \wedge \text{ActiveLowEnergyReadout} \wedge \text{SMFieldInventory}^{\text{cert}} \\ \wedge \text{LowEnergyQuotient}_{\mathcal{W}_{\text{low}}} \implies \text{TargetFix}(\mathfrak{D}_{\nu,0}^{\text{LE}}) = 1. \end{aligned} \quad (14)$$

Thus $\mathfrak{D}_{\nu,0}^{\text{LE}}$ is induced before branch evaluation by the admitted witness family, certified carrier inventory, and readout quotient; it is not selected by preference for the Weinberg branch.

Here $\text{SMFieldInventory}^{\text{cert}}$ is not an empirical convention by itself. It denotes the currently source-certified light field-carrier and bridge inventory supplied by the Standard-Model reconstruction, charge/anomaly closure, Higgs-bridge, Yukawa-compatibility, and realized-physical-interior dependencies. A written field outside that inventory enters only by the carrier-certification and target-update protocol.

Proof. The PMNS mass-driven witness fixes an active-neutrino propagation readout and, by Theorem 4.2, the mass-role target that the readout interrogates. R6 and the Standard-Model source lock fix the carrier inventory actually certified in the current source state; a merely writable new field does not enter the same target without a carrier certificate. The family $\mathcal{W}_{\text{low}}$ fixes which distinctions survive the low-energy readout, so richer descriptions with identical projections are one target-level quotient class rather than additional same-target occupants. These conditions jointly determine the

minimal fixed comparison domain that preserves the admitted readout, carrier load, mass role, and quotient identity in the current corpus snapshot. That domain is exactly $\mathfrak{D}_{\nu,0}^{\text{LE}}$. A future certified carrier/protector or surviving-witness package would be evaluated by $\mathfrak{D}_{\nu,+}^{\text{LE}}$ and would define an updated target state. No exact-character branch has yet been selected: target fixation determines what counts as a same-target competitor, while branch elimination occurs only through the later operator, protector, audit, and selection theorems. $\square$

Remark 4.6 (Target fixation is not branch selection). Theorem 4.5 does not infer the Weinberg occupant or reject exact Dirac. It fixes the current evaluation domain $\mathfrak{D}_{\nu,0}^{\text{LE}}$ from witness and carrier data. Theorem 8.5 states the independent burden for exact Dirac, Definition 2.7 fixes the audit acceptance criterion, and Theorems 12.4 and 12.5 perform the later certified same-target branch deletion and selection.

5 Declared comparison envelopes

5.1 Field and charge convention

Use the Standard-Model gauge product

$$G_{\text{SM}} = SU(3)_c \times SU(2)_L \times U(1)_Y, \quad Q = T_3 + Y. \quad (15)$$

For each family,

$$L_i \sim (\mathbf{1}, \mathbf{2}, -\tfrac{1}{2}), \quad e_{Ri} \sim (\mathbf{1}, \mathbf{1}, -1), \quad H \sim (\mathbf{1}, \mathbf{2}, +\tfrac{1}{2}), \quad (16)$$

and $\tilde{H} = i\sigma_2 H^*$ transforms as $(\mathbf{1}, \mathbf{2}, -\tfrac{1}{2})$. These charges agree with the rigid charged chiral skeleton of the companion charge and Standard-Model sources [10, 12].

Remark 5.1 (Spinor and electroweak notation). The renormalizable Dirac Yukawa term is written consistently in four-component notation as $\bar{L} \tilde{H} N_R$. The Weinberg class is written in left-handed notation using $L \cdot H \equiv \epsilon_{ab} L^a H^b$, with spinor indices contracted by C . These are notation choices for different operator types, not a mixed hypercharge convention.

5.2 Envelope definitions

Definition 5.2 (Minimal renormalizable envelope $\mathcal{E}_0$). $\mathcal{E}_0$ contains the minimal Standard-Model fields, the one-doublet bridge skin, local Lorentz-invariant operators of canonical dimension $d \leq 4$, and no gauge-singlet right-handed neutrino.

Definition 5.3 (Standard-Model-field effective envelope $\mathcal{E}_5$). $\mathcal{E}_5$ contains the same light fields and gauge structure as $\mathcal{E}_0$ but admits local nonredundant operators through $d = 5$.

Definition 5.4 (Renormalizable singlet envelope $\mathcal{E}_{\nu_R}$). $\mathcal{E}_{\nu_R}$ augments $\mathcal{E}_0$ by n_R right-handed gauge-singlet neutrino fields $N_{Ra} \sim (\mathbf{1}, \mathbf{1}, 0)$ while retaining $d \leq 4$ operators. Their charge-conjugates appear only when the neutral-fermion mass matrix is written in a left-chiral basis.

Definition 5.5 (Richer-scope category $\mathcal{E}_+$). $\mathcal{E}_+$ indexes any declared extension adding non-singlet heavy fermions, scalar multiplets, new gauge factors, additional bridge fields, radiative sectors, nonlocal structure, extra dimensions, or other standing-bearing roles absent from the finite envelopes above. It is an open category governed here by a low-energy quotient rule; the symbol does not assert an enumeration of all ultraviolet theories.

Remark 5.6 (Why $\mathcal{E}_5$ and $\mathcal{E}_{\nu_R}$ are incomparable rather than sequential). $\mathcal{E}_5$ changes the operator-dimension ceiling while preserving the light field load. $\mathcal{E}_{\nu_R}$ changes the field load while preserving renormalizability. Neither is automatically a refinement of the other. A combined singlet effective envelope could be declared, but it would be a separate comparison object and is not needed for the principal exhaustion.

5.3 Candidate operator classes

The candidate low-energy mass-response classes are

$$\begin{aligned}
\mathcal{C}_0 &: \text{no neutrino mass carrier,} \\
\mathcal{C}_D &: \bar{L} \tilde{H} N_R \quad (\text{Dirac Yukawa}), \\
\mathcal{C}_W &: (L \cdot H)^T C (L \cdot H) / \Lambda \quad (\text{Weinberg/Majorana}), \\
\mathcal{C}_R &: \bar{N}_R^c N_R \quad (\text{singlet Majorana}), \\
\mathcal{C}_{\text{mix}} &: \text{simultaneous Dirac and Majorana blocks,} \\
\mathcal{C}_{\text{pD}} &: \text{pseudo-Dirac regime with parametrically small Majorana blocks,} \\
\mathcal{C}_+ &: \text{richer-scope completions and their low-energy residues.}
\end{aligned} \tag{17}$$

Here and below, schematic products suppress spinor and $SU(2)$ contractions; explicit forms are given in Appendix A.

6 The minimal renormalizable obstruction

6.1 Operator enumeration in $\mathcal{E}_0$

A Lorentz-invariant fermion mass term couples opposite chiralities in a Dirac bilinear or couples a chiral field to its charge conjugate in a Majorana bilinear. In $\mathcal{E}_0$, the active neutrino occurs only inside L_i . Three candidate repairs must therefore be tested.

Direct Dirac term. A term $\overline{\nu_{Li}} \nu_{Rj}$ requires a right-handed singlet ν_R , which is absent by definition.

Direct left-handed Majorana term. The bilinear $L_i^T C \epsilon L_j$ carries hypercharge -1 . Although the antisymmetric $SU(2)$ contraction is a singlet, the bilinear is not neutral under $U(1)_Y$ and cannot appear as a gauge-invariant dimension-three mass term.

Bridge-assisted renormalizable term. A single Higgs insertion cannot convert two left-handed lepton doublets into a Lorentz-scalar, gauge-neutral dimension-four operator. Two Higgs insertions are required, which raises the operator to dimension five.

Theorem 6.1 (NMC-T2: minimal renormalizable active-neutrino mass obstruction). *Within $\mathcal{E}_0$, no active-neutrino mass-response operator is available:*

$$\text{Avail}_{\mathcal{E}_0}(\mathcal{C}_D) = \text{Avail}_{\mathcal{E}_0}(\mathcal{C}_W) = \text{Avail}_{\mathcal{E}_0}(\mathcal{C}_R) = 0. \tag{18}$$

Proof. The Dirac class requires a field absent from the envelope. A direct left-handed Majorana bilinear fails hypercharge neutrality. The gauge-neutral bridge-assisted left-handed Majorana contraction requires two lepton doublets and two Higgs doublets and therefore has canonical dimension five, outside the $d \leq 4$ ceiling. Singlet Majorana terms are unavailable because no singlet neutrino is present. These possibilities exhaust local quadratic neutrino mass responses built from the declared load. The conclusion agrees with the companion Standard-Model closure, whose declared minimal renormalizable load contains no additional gauge-invariant operator and treats the neutrino load choice as external to that load [12]. $\square$

Corollary 6.2 (NMC-C3: oscillation excludes $\mathcal{E}_0$ as a complete realized envelope). *If $W_{\text{osc}}^{m,\text{PMNS}}$ is admitted, $\mathcal{E}_0$ cannot be the complete envelope of the realized neutrino sector.*

Proof. Corollary NMC-C2 requires nontrivial mass-response occupancy somewhere, whereas Theorem 6.1 supplies no available carrier in $\mathcal{E}_0$. The contradiction is resolved only by changing field content, operator dimension, or another declared envelope component. $\square$

Remark 6.3 (Nature of the obstruction). This is not a statement that a Yukawa coefficient happens to be small. It is an operator-slot obstruction. The MGN4 and RPI sources explicitly distinguish Yukawa realizability from numerical Yukawa values and retain an unresolved neutrino status variable rather than silently filling the slot [8, 20].

7 The dimension-five Standard-Model-field envelope

7.1 The Weinberg class

In $\mathcal{E}_5$, define the electroweak singlet spinor

$$L_i \cdot H \equiv \epsilon_{ab} L_i^a H^b, \quad \epsilon = i\sigma_2, \quad (19)$$

and the Weinberg class

$$\mathcal{O}_5^{ij} = (L_i \cdot H)^T C (L_j \cdot H), \quad \mathcal{L}_5 = \frac{1}{2\Lambda} \kappa_{ij} \mathcal{O}_5^{ij} + \text{h.c.} \quad (20)$$

The normalization and equivalent index-contraction skins are not standing-bearing. Fermi statistics and the contractions imply that the physically relevant flavor coefficient may be represented by a complex symmetric matrix κ . After the neutral bridge component acquires its ordinary broken-phase expectation value, the operator produces a symmetric active-neutrino mass matrix

$$m_L \propto \kappa \frac{v^2}{\Lambda}. \quad (21)$$

The proportionality factor depends on the convention for H and v and carries no role in the classification.

7.2 Uniqueness through dimension five

Remark 7.1 (Operator-basis convention). The uniqueness claim is made for local Lorentz- and G_{SM} -invariant operators built from the declared Standard-Model light fields through canonical

dimension five. Operators are quotiented by integration by parts, algebraic index identities, lawful field redefinitions and equations of motion at the declared EFT order, Hermitian conjugation, flavor-basis redescription, and presentation-only normalization. The target is specifically an operator that produces an active-neutrino mass after bridge activation. Under this basis convention the surviving class is $[\mathcal{O}_5]$.

Theorem 7.2 (NMC-T3: dimension-five active-neutrino operator uniqueness). *Within $\mathcal{E}_5$, the Weinberg class $[\mathcal{O}_5]$ is the unique nonredundant local operator class through dimension five that produces an active-neutrino mass after bridge activation. Equivalently,*

$$|\{C : \text{Avail}_{\mathcal{E}_5}(C) = 1, C \text{ supplies active-neutrino mass}\} / \sim| = 1. \quad (22)$$

Proof. The field load contains no N_R , so a Dirac Yukawa term is unavailable. A mass-generating operator must contain two lepton doublets to form a Lorentz bilinear. Their total hypercharge is -1 , requiring fields of total hypercharge $+1$. The minimal available compensation is two Higgs doublets, each with $Y = +1/2$. The resulting canonical dimension is $2(3/2) + 2(1) = 5$. The $SU(2)$ indices may be expressed in equivalent contraction skins related by identities and integration-by-parts or field-redefinition reductions; they descend to the same Weinberg operator class. No operator with fewer fields satisfies Lorentz and gauge invariance, and any additional derivative or field strength raises the dimension. This is the conventional dimension-five uniqueness result of Standard-Model effective field theory [24, 25]. $\square$

Corollary 7.3 (NMC-C4: envelope-relative Majorana character). *Any occupied active-neutrino mass response in $\mathcal{E}_5$ has Majorana character and violates the ordinary lepton-number assignment by two units.*

Proof. The unique class couples two left-handed lepton fields and, after bridge activation, yields a symmetric same-chirality mass matrix. No independent right-handed partner exists in the envelope. The resulting massive neutral fields are represented by Majorana mass eigenstates. The operator changes the conventional lepton-number charge by $\Delta L = 2$. $\square$

7.3 Availability is not occupancy

Theorem 7.4 (NMC-T4: operator availability does not alone imply occupancy). *The uniqueness of $[\mathcal{O}_5]$ in $\mathcal{E}_5$ does not, by itself, imply $\kappa \neq 0$, determine $\text{rank}(\kappa)$, fix its eigenvalues, or identify Λ .*

Proof. The zero coefficient is compatible with the operator ledger. Operator enumeration establishes that a class may appear; it does not assign a nonzero coefficient. Even after nonzero occupancy is certified, the CNF6/MGN5 non-promotion wall separates support, rank, quotient spectrum, scale, and calibrated mass values [5, 9]. The bridge source likewise prevents ν or a Higgs-like presentation from automatically fixing a neutrino-specific scale [7]. $\square$

Corollary 7.5 (NMC-C5: conditional Weinberg occupancy with spectral support). *If $\mathcal{E}_5$ is declared complete for the realized light-neutrino sector and $W_{\text{osc}}^{m, \text{PMNS}}$ is admitted, then the Weinberg class is occupied by a coefficient structure whose induced mass matrix has nontrivial quotient-spectrum support:*

$$\text{Occ}_{\mathcal{E}_5}^{\text{cert}}([\mathcal{O}_5]) = 1, \quad \left| \text{Spec}(m_L^\dagger m_L) \right|_{W_{\text{osc}}^{m, \text{PMNS}}} \geq 2, \quad \kappa \neq 0, \quad (23)$$

where the middle condition means that at least two squared-mass eigenvalues distinguished by the boundary mixing support are unequal.

Proof. Corollary NMC-C2 requires a nontrivial mass-response carrier. Theorem 7.2 supplies exactly one available class in the complete envelope, so the required occupancy must factor through that class. Occupancy by a coefficient proportional to a fully degenerate squared-mass spectrum would not reproduce $W_{\text{osc}}^{m, \text{PMNS}}$; the induced $m_L^\dagger m_L$ must therefore contain the nontrivial quotient-spectrum distinction read by the oscillation kernel. Nonzero κ follows, but is weaker than the full witness condition. The conclusion fixes neither coefficient values nor ultraviolet origin. $\square$

Source note 7.6 (Correction of a common overcautious statement). “Availability is not occupancy” remains essential, but once the oscillation mass witness and the completeness of $\mathcal{E}_5$ are both admitted, occupancy is conditionally forced. The constraint-grade conclusion is therefore not merely “Weinberg available”; it is “Weinberg occupied with the required quotient-spectrum support, conditional on the $\mathcal{E}_5$ envelope.” The global problem remains open because the source stack does not select $\mathcal{E}_5$ over competing envelopes.

7.4 What remains free in $\mathcal{E}_5$

Even after conditional occupancy, the following remain unselected:

- the overall coefficient scale $\|\kappa\|/\Lambda$;
- the rank and nullity of m_L ;
- the absolute mass offset and lightest mass;
- the mass ordering and normalized mass shape unless fixed by independent source theorems;
- the Majorana phase residues;
- the ultraviolet completion responsible for $\mathcal{O}_5$.

The output is a character and operator-class selection inside one envelope, not a numerical mass solution.

8 The renormalizable singlet extension

8.1 Dirac and singlet-Majorana operators

Add n_R gauge-singlet fields $N_{Ra} \sim (\mathbf{1}, \mathbf{1}, 0)$. The charge-lattice source explicitly records such a field as anomaly-inert in the charged hypercharge theorem [10]. The renormalizable neutrino terms are

$$\mathcal{L}_\nu^{(4)} = -\bar{L}_i \tilde{H} (Y_\nu)_{ia} N_{Ra} - \frac{1}{2} \overline{N_{Ra}^c} (M_R)_{ab} N_{Rb} + \text{h.c.} \quad (24)$$

After bridge activation, $m_D \propto Y_\nu v$.

Proposition 8.1 (NMC-P1: Dirac availability in $\mathcal{E}_{\nu_R}$). *Within $\mathcal{E}_{\nu_R}$, the Dirac Yukawa class is gauge- and Lorentz-admissible. The singlet addition therefore removes the Dirac operator-slot obstruction of $\mathcal{E}_0$.*

Proof. $\bar{L}$ has hypercharge $+1/2$, $\tilde{H}$ has hypercharge $-1/2$, and N_R has hypercharge zero, so the product is neutral. The doublet indices contract to a singlet, and the fermion bilinear is Lorentz

invariant. The MGN4 framework types this as coupling-slot availability over an admitted bridge carrier; it does not fix Y_ν numerically [8]. $\square$

Proposition 8.2 (NMC-P2: singlet admission leaves both bilinear classes available). *Within $\mathcal{E}_{\nu_R}$, gauge and Lorentz invariance admit both the Dirac Yukawa class and a singlet Majorana class.*

Proof. Because N_R is a complete gauge singlet, the bilinear $\overline{N}_R^c N_R$ is gauge neutral and has dimension three. No Standard-Model gauge constraint sets M_R to zero. Thus the same field extension that makes m_D available also makes M_R available. $\square$

8.2 Generic mass matrix

In the left-chiral basis

$$n_L = (\nu_L, N_R^c)^T, \quad (25)$$

the general neutral-fermion mass matrix, allowing also a left-Majorana block if a dimension-five term is later admitted, is

$$\mathcal{M}_\nu = \begin{pmatrix} m_L & m_D \\ m_D^T & M_R \end{pmatrix}, \quad \mathcal{L}_m = -\frac{1}{2} \overline{n}_L \mathcal{M}_\nu n_L^c + \text{h.c.} \quad (26)$$

Within strictly renormalizable $\mathcal{E}_{\nu_R}$, $m_L = 0$ while m_D and M_R are available.

Lemma 8.3 (NMC-L2: neutral-spinor exact-character normal form). *For a neutral fermionic mass-role occupant in the declared Lorentz, gauge, carrier, and quotient scope, the exact Dirac–Majorana axis is equivalent to preservation or failure of a nontrivial quotient-stable additive particle/antiparticle persistence relation Λ_ν . A Dirac bilinear can preserve that relation when an exact protector exists; an occupied same-chirality Majorana bilinear violates it. Mixed and pseudo-Dirac systems refine the violated branch by block profile, rank, splitting, and readout resolution; they do not supply a third exact standing value.*

Proof. The candidate exact-character coordinates can be eliminated independently of the desired conclusion. Electric charge cannot perform the role because the target carrier is electrically neutral. A mass eigenvalue, mass ordering, or splitting selects spectral data, not whether the mass response preserves an additive particle/antiparticle relation. PMNS coordinates and propagation phases belong to the flavor-propagation quotient, while diagonalization and field rephasings are basis redescrptions. An ultraviolet mechanism name identifies an origin route and quotient-collapses when no character-sensitive witness survives.

The remaining exact distinction is whether the occupied neutral-fermion mass response preserves a nontrivial additive persistence relation between particle and antiparticle carrier roles. A neutral Dirac field uses two independent chiral carrier roles joined by an off-diagonal mass response and admits such a relation when every same-chirality mass term is protectively excluded. A Majorana bilinear joins a chiral field to its charge conjugate and changes the additive relation by two units, so exact preservation fails. If both bilinear types are occupied, or if nonzero Majorana splittings pair approximately into a Dirac response, the exact relation still fails even though a finite-resolution readout may be approximately Dirac-like. Gauge, basis, phase, and permutation redescrptions do not alter which bilinear classes are occupied. The exact axis is therefore a quotient-stable structural relation, not a definition chosen to favor one branch. $\square$

Table 4: Elimination of alternative exact-character axes.

Candidate axis	Why it fails or reduces
Electric charge	The active-neutrino target is electrically neutral; electric charge does not distinguish Dirac from Majorana mass character.
Mass eigenvalue, ordering, or splitting	These are scale and spectrum coordinates that can be shared by exact-Dirac and Majorana-compatible realizations.
PMNS mixing coordinates	They belong to the flavor-propagation quotient and do not determine the occupied bilinear class.
Matrix diagonalization or field phase	These are admissible basis normal forms and redescrptions, not exact-character selectors.
Majorana column phase	It is character-sensitive only through an independently admitted violating channel; it is not a separate exact-character coordinate in standard oscillation.
Ultraviolet mechanism label	It selects an origin route and collapses to the low-energy residue when no witness in $\mathcal{W}_{\text{low}}$ survives.
Additive particle/antiparticle persistence relation	This is the surviving exact-character axis: exact preservation supports Dirac standing, while occupied same-chirality mass response violates it.

In conventional spinor language, this is the AASC role-level analogue of the familiar distinction that exact Dirac character requires a conserved particle/antiparticle charge distinction, whereas an occupied Majorana mass term fails that distinction at the mass-role level. The AASC formulation is stricter: the distinction must be standing-bearing in the declared target, not merely named by notation.

Definition 8.4 (Protective standing condition). *A protective condition $\mathfrak{P}_D$ for pure Dirac character must satisfy all of the following:*

1. *it forbids M_R , m_L , and every other Majorana operator admitted by the declared envelope;*
2. *it is not merely the coefficient stipulation $M_R = m_L = 0$;*
3. *it survives the admitted quantum, renormalization, matching, and transport operations;*
4. *it has standing independently of the desired Dirac conclusion; and*
5. *if it adds a gauge factor, field, discrete structure, or boundary rule, that load is declared as richer scope rather than hidden inside $\mathcal{E}_{\nu_R}$.*

Table 5: Protective-condition audit for an exact Dirac branch.

Candidate protector	Forbids	Majorana terms?	Standing status	AASC disposition
$M_R = m_L = 0$ by coefficient choice	At the displayed order	No	independent protector	Forbidden from chosen zero to protected zero.
Imposed global lepton number	Formally, within the stated model	Not automatic	standing-bearing	Admissible model assumption; selection requires stability and standing certification.
Unbroken anomaly-closed gauged $B - L$	Conditionally	Potentially	standing-bearing	Richer scope unless already admitted; symmetry breaking pattern must remain compatible with exact protection.
Discrete gauge or remnant symmetry	Conditionally	Potentially	standing-bearing	Requires an all-operator certificate in the declared envelope.
Accidental symmetry at finite operator order	Only through the truncation	Incomplete	standing-bearing	Does not establish exact Dirac standing to all admitted orders.
Null $\Delta L = 2$ searches	No	Empirical	non-observation only	Nonselector for an exact all-operator zero.

Theorem 8.5 (NMC-T5: exact-Dirac carrier/protector burden theorem). *Inside any declared envelope, exact Dirac character requires*

$$\text{Occ}_{\mathcal{E}}^{\text{cert}}(\mathcal{C}_D) = 1, \quad \text{CarrierPkg}_{\nu_R}^{\text{cert}} = 1, \quad \text{ProtectorPkg}_{\Delta L}^{\text{cert}} = 1, \quad \text{Stand}_{\Delta L, \mathcal{E}}(\mu_D(\Pi_\nu)) = 1. \quad (27)$$

The protector must satisfy

$$\forall \mathcal{O}_{\Delta L=2} \in \mathcal{E}_{\text{declared}}, \quad \text{Avail}_{\mathcal{E}}(\mathcal{O}_{\Delta L=2}) = 0 \quad \text{or} \quad \mathfrak{P}_D \text{ protectively forbids } \mathcal{O}_{\Delta L=2}. \quad (28)$$

If the admitted effective envelope is unbounded, finite-order inspection is insufficient: $\mathfrak{P}_D$ must be a standing-bearing theorem covering the admitted operator tower. A written N_R , a displayed Majorana zero, or a finite null search is not the required package.

Proof. The singlet field permits both the Dirac Yukawa operator and singlet Majorana bilinears; higher admitted orders can also contain $\Delta L = 2$ terms. Exact Dirac character therefore requires an operative Dirac-completing carrier, not merely a symbol, and an independently certified obstruction preserving Λ_ν against every admitted violating route. R6 excludes hidden, default, trace-only, and bookkeeping carriers as field-role occupants. Persistence-regime realization excludes abstract-only protection. A coefficient zero supplies no reason for persistence under matching, renormalization, transport, or scope extension. Conversely, an occupied Dirac block plus a certified carrier and all-operator protector preserves the exact additive distinction. The condition is envelope-relative; a richer-scope package must be declared as a new target. $\square$

Corollary 8.6 (NMC-C6: generic singlet survivor set). *Without a protective standing condition, the renormalizable singlet envelope admits at least the following regimes:*

$$\mathcal{S}(\mathcal{E}_{\nu_R}) = \{\text{Dirac coefficient branch, SingletMajorana, Mixed, PseudoDirac}\}, \quad (29)$$

where the unprotected pure-Dirac branch is available but not structurally selected.

Definition 8.7 (Pseudo-Dirac status relative to a readout). *A neutral-fermion pair is pseudo-Dirac relative to readout $\mathcal{R}$ when it consists of two Majorana eigenstates whose quotient-stable average mass and mixing combine into an approximate Dirac response, while a nonzero pair splitting δm^2 remains below or near the declared resolution of $\mathcal{R}$. The status is therefore readout-relative and approximate; exact Dirac standing requires the protected vanishing of every Majorana block.*

Remark 8.8 (No matrix-norm shortcut). The raw inequality $\|m_L\|, \|M_R\| \ll \|m_D\|$ may be useful in a chosen representation, but it is not by itself the AASC definition. The standing object is the quotient-stable paired spectrum, approximate conserved-charge response, and declared resolution scale [5].

9 Mass-matrix quotient and character classes

9.1 Basis-redescription discipline

For n_L active and n_R singlet fields, admissible flavor-basis transformations act as

$$\nu_L \mapsto V_L \nu_L, \quad N_R \mapsto V_R N_R, \quad (30)$$

with unitary V_L, V_R compatible with the charged-current and charged-lepton comparison ledger. The mass blocks transform as

$$m_D \mapsto V_L^\dagger m_D V_R, \quad m_L \mapsto V_L^T m_L V_L, \quad M_R \mapsto V_R^T M_R V_R. \quad (31)$$

Raw entries are therefore presentation data. The CNF6 and MGN5 sources require mass claims to descend to quotient-stable support, rank, nullity, degeneracy, singular-value, or spectral objects [5, 9].

Definition 9.1 (Neutrino mass-character quotient). *Two neutrino mass carriers are character-equivalent when they are related by admissible flavor rotations, field rephasings, permutation of physically indistinguishable eigenstates, and low-energy field redefinitions while preserving (i) the existence or absence of an exact conserved lepton-number-type charge, (ii) the number and rank of Majorana blocks, and (iii) all declared boundary observables.*

Proposition 9.2 (NMC-P3: diagonalization is not character selection). *A diagonal matrix presentation does not select Dirac or Majorana character. Character is fixed by the field-identification and conserved-charge structure of the mass carrier, not by whether a representative matrix has been diagonalized.*

Proof. Every complex Dirac mass matrix admits a singular-value decomposition, while every complex symmetric Majorana mass matrix admits a Takagi factorization. Both yield nonnegative diagonal entries, but the field content and allowed charge assignment differ. CNF6 explicitly prohibits promotion of a preferred diagonal representative into a physical selector [5]. $\square$

9.2 Normal forms

Dirac class. If $m_L = M_R = 0$, a biunitary transformation gives

$$U_L^\dagger m_D U_R = \text{diag}(m_1, \dots, m_r, 0, \dots), \quad (32)$$

and each nonzero singular value pairs two Weyl fields into a Dirac fermion.

Majorana class. If a symmetric mass matrix $M = M^T$ acts on a set of left-chiral neutral fields, a Takagi factorization gives

$$U^T M U = \text{diag}(m_1, \dots, m_r, 0, \dots), \quad m_i \geq 0, \quad (33)$$

with self-conjugate mass eigenfields after the usual construction.

Mixed class. For simultaneous m_D and Majorana blocks, the full symmetric matrix in Eq. (26) is Takagi-diagonalized. The eigenstates are generically Majorana even when some pairs form approximate Dirac combinations.

Pseudo-Dirac class. Relative to a declared readout, two Majorana eigenstates may form an approximate Dirac pair when their average mass and mixing reproduce a Dirac-like response while a nonzero pair splitting remains below or near the readout resolution. The approximation does not restore exact Dirac standing unless every Majorana block vanishes under a protective condition.

Proposition 9.3 (NMC-P4: character-class invariance). *Dirac, exact Majorana, mixed, and pseudo-Dirac statuses are invariant under admissible flavor-basis redescription when defined by the conserved-charge and quotient-spectral conditions above.*

Proof. Unitary changes of basis preserve the existence of a global continuous charge commuting with the mass term, preserve matrix rank, and preserve Takagi or singular values. They can move phases and texture entries but cannot turn a nonzero Majorana block into an exactly charge-preserving Dirac theory, or vice versa, without changing the operator content. Approximate pseudo-Dirac status additionally depends on quotient-stable relative splittings; it is therefore invariant once the scale and resolution ledger is declared. $\square$

9.3 Rank consequences of the oscillation witness

Proposition 9.4 (NMC-P5: rank lower bound from two independent splitting roles). *If the realized light-neutrino sector contains three propagation states and two independent nonzero mass-squared splitting witnesses, then the light mass matrix has rank at least two. One exactly massless light state remains compatible with that witness set.*

Proof. Two independent nonzero squared-mass differences require at least three squared eigenvalues not all equal and at least two nonzero masses. A rank-one light mass matrix would have at most one nonzero mass and only one independent nonzero squared-mass scale relative to two zero eigenvalues. Rank two permits one zero eigenvalue and two nonzero eigenvalues, yielding two independent differences. The statement is structural and does not select their values or ordering, in accord with CNF6 and NNR3 [5, 3]. $\square$

Remark 9.5 (Singlet cardinality). In the minimal type-I limit with $m_L = 0$ and nonsingular heavy M_R , the effective light matrix has $\text{rank}(m_\nu) \leq \text{rank}(m_D) \leq n_R$. Thus two independent light splittings require $n_R \geq 2$ in that restricted route, while $n_R = 2$ permits one massless light state. This is a conditional rank statement, not a derivation of the number of singlets.

10 Richer-scope origins and the low-energy quotient rule

10.1 Origin is not identical to character

A low-energy Majorana mass may arise from several heavy sectors. Conversely, a Dirac mass may arise from a direct tiny Yukawa coupling or from a richer-scope Dirac seesaw or radiative mechanism. The low-energy character target and the ultraviolet-origin target must therefore be separately typed.

Definition 10.1 (Declared low-energy witness family). $\mathcal{W}_{\text{low}}$ is the complete witness family declared for the low-energy mass-character target. It may include the light-neutrino operator class, coefficient tensor within resolution, quotient mixing structure, nonunitarity, correlated effective operators, and any other explicitly admitted record. It does not silently include collider, cosmological, or heavy-state channels unless they are named in the target.

Definition 10.2 (Low-energy mass-character equivalence). Two ultraviolet mechanisms X and Y are equivalent relative to $\mathcal{W}_{\text{low}}$ when $\pi_{\text{low}}(X) = \pi_{\text{low}}(Y)$ and

$$\forall W \in \mathcal{W}_{\text{low}}, \quad W(X) = W(Y). \quad (34)$$

They may remain distinct ultraviolet theories while being one answer at the declared low-energy character target.

Source note 10.3 (No unbounded $\mathcal{E}_+$ exhaustion). $\mathcal{E}_+$ is an open richer-scope category, not a finite model list. This paper does not claim to enumerate every ultraviolet construction. Its complete statement is a quotient rule: any declared richer-scope route is distinct at the low-energy character target only if some standing-bearing difference survives the projection π_{low} to the declared readout.

10.2 Representative routes

Table 6: Representative, non-exhaustive routes and their AASC typing.

Route	Added bearing load	standing-load	Low-energy residue	mass	Audit status
Type-I seesaw	Gauge-singlet fermions N_R	heavy	Weinberg heavy singlets integrated out	class when are	Same scope as $\mathcal{E}_{\nu_R}$; richer than $\mathcal{E}_0/\mathcal{E}_5$ as a UV explanation
Type-II seesaw	Heavy scalar triplet with appropriate hypercharge	$SU(2)_L$ with appropriate	Weinberg class plus possible scalar-sector residues	when residues	Richer scope; distinguishable only if non-Weinberg witnesses survive

Route	Added standing-bearing load	Low-energy residue	mass	Audit status
Type-III see-saw	Heavy fermion $SU(2)_L$ triplet	Weinberg gauge-interaction residues	class plus	Richer scope; representation ledger changes
Radiative Majorana route	Additional fields/symmetries and loop structure	Usually Weinberg class at low energy, possibly higher operators		Richer scope; loop order is not itself a low-energy character witness
Inverse/linear seesaw	Additional singlets and approximate lepton-number structure	Mixed/pseudo-Dirac Majorana light sector	or	Richer than minimal $\mathcal{E}_{\nu_R}$ unless all roles declared there
Protected Dirac route	Exact global, gauge, or discrete protector plus partner fields as required	Dirac mass operator with all Majorana terms forbidden	Protector must be source-certified; often richer scope	
Higher-dimensional route	Operators with $d > 5$ or nonminimal bridge content	Dirac or Majorana depending on fields and symmetry	de-	Richer operator envelope

The historical tree-level seesaw routes and their effective Weinberg residue are standard [26, 27, 28, 29, 30, 32].

Theorem 10.4 (NMC-T6: low-energy ultraviolet quotient theorem). *Let X and Y be any two declared ultraviolet mechanisms. If*

$$\pi_{\text{low}}(X) = \pi_{\text{low}}(Y), \quad \forall W \in \mathcal{W}_{\text{low}}, \quad W(X) = W(Y), \quad (35)$$

then $X \sim_{\text{low}} Y$ for the neutrino mass-character target, even though they remain distinct ultraviolet theories.

Proof. The low-energy readout factors through the matched effective action. By hypothesis, every object read by the declared witness family agrees after admissible field redefinition and running/matching transport. The heavy-field identity is absent from the accessible quotient, and no residual operator distinguishes the routes. By standing-bearing sameness and Criterion 2.5, counting the two mechanisms as distinct low-energy character survivors would duplicate a quotient-equivalent class. A higher-energy witness, correlated operator, collider signature, flavor-violation pattern, or cosmological record may enlarge the readout and separate them; the theorem does not enumerate or select the ultraviolet mechanisms themselves. $\square$

Corollary 10.5 (NMC-C7: ultraviolet multiplicity does not multiply the low-energy character endpoint). *Counting every type-I, type-II, type-III, or radiative realization as a distinct answer to “Dirac or Majorana?” overcounts the character survivor set. Mechanism multiplicity belongs to the origin ledger unless a distinguishing witness survives.*

Principle 10.6 (No decoupling overread). *The fact that a heavy route matches to the Weinberg class does not prove that the heavy route exists. Matching supplies an equivalence map conditional on the ultraviolet model; it is not an occupancy theorem for that model.*

11 Scale, shape, and anchor firewalls

11.1 Difference space is not mass shape

Let

$$\mathbf{s} = (m_1^2, m_2^2, m_3^2). \quad (36)$$

Oscillation reads the projection of $\mathbf{s}$ to the quotient

$$\mathbf{s} \sim \mathbf{s} + c(1, 1, 1), \quad (37)$$

not the absolute triple. NNR3 proves that a splitting-witness quotient, a squared-eigenvalue shape, a mass-eigenvalue shape, and an absolute normalized mass shape are distinct targets; no lawful quotient collapses them without an offset and normalization bridge [3].

Proposition 11.1 (NMC-P6: scale-and-shape separation). *The oscillation witness and an operator-character classification do not determine the absolute neutrino mass scale. In particular, neither*

$$[\mathcal{O}_5] \text{ occupied} \quad \text{nor} \quad \text{Dirac selected} \quad (38)$$

fixes the common mass offset, the lightest mass, or the suppression scale of the underlying operator.

Proof. Oscillation is invariant under a common squared-mass translation. Operator character fixes the bilinear and symmetry type but leaves coefficient magnitudes. In $\mathcal{E}_5$, $m_L \propto \kappa v^2/\Lambda$ contains an undetermined coefficient-scale combination. In a Dirac branch, $m_D \propto Y_\nu v$ contains undetermined Yukawa singular values. CNF6 forbids spectral-value promotion; MGN3 forbids automatic bridge-scale promotion; NNR3 blocks difference-to-shape promotion [5, 7, 3]. $\square$

11.2 Anchor admissibility

NNR0.8 defines an anchor as a source-certified datum that adds only a nonselective scale coordinate to an already closed dimensionless target. It proves both “no anchor before shape” and “anchor cannot select residue” [4].

Corollary 11.2 (NMC-C8: no character selection by scale anchor). *A beta-decay scale, cosmological mass sum, bridge scale, or other absolute datum cannot by itself choose Dirac versus Majorana character unless the datum carries an independent character-sensitive witness.*

Proof. A lawful anchor normalizes an already fixed target and may not alter non-scale role occupation. Dirac and Majorana sectors can share the same absolute mass spectrum. A scale-only datum therefore does not separate them. $\square$

Corollary 11.3 (NMC-C9: no Higgs-scale shortcut). *The electroweak bridge scale does not determine the neutrino mass scale without a neutrino-specific relation fixing Y_ν or κ/Λ .*

Proof. The bridge supplies the conversion factor appearing in a mass term but not the dimensionless Yukawa value or the effective suppression coefficient. This is the explicit bridge-scale boundary of MGN3 and NNR0.8 [7, 4]. $\square$

11.3 Numerical dependency chronology and firewall

NNR3 and NNR0.8 are obstruction and anchor-readiness papers. The later RPI synthesis reports a tightly constrained neutrino ratio and one-anchor display, but the downstream NNR proofs supporting that synthesis are not included in the source lock used here. Reuse therefore stops at the results directly supported by the archived sources. Nothing in the character theorem depends on whether the later numerical arc is ultimately admitted, because a mass ratio or absolute scale is not a Dirac/Majorana selector by itself.

12 Selector exhaustion and the certified low-energy endpoint

12.1 Neutral-spinor exact-character normal form

Lemma 8.3 supplies the structural exact-character axis used below. It is proved in the singlet-envelope analysis before selector exhaustion, so the selector result does not define its own comparison coordinate.

12.2 Exhaustion of proposed exact-character selectors

Table 7: Exhaustion of candidate exact-character selectors.

Candidate	What it actually selects	Constraint disposition
Mass-driven oscillation	Spectral nontriviality and invariant mixing support	Fixes the mass-role target, not character.
Absolute mass datum or mass sum	Scale coordinate under a model/transport map	Dirac and Majorana-compatible theories can share it.
PMNS coordinates or phases	Flavor-propagation quotient	Majorana column phases cancel from standard oscillation.
Matrix diagonalization	Basis normal form and quotient spectrum	Redescription does not change bilinear character.
Weinberg availability	Unique $d = 5$ active-mass class in $\mathcal{E}_5$	Selects character only after the low-energy envelope is fixed and the class is occupied.
Admission of N_R	Availability of a Dirac Yukawa term	Also admits Majorana-compatible terms and changes the field load.
Displayed coefficient zero	A chosen parameter point	Not an all-operator standing obstruction.
Naturalness or minimality	Preference/ranking	No admissibility-bearing selector authority.
Ultraviolet model name	Origin route	Quotient-collapses when no witness in $\mathcal{W}_{\text{low}}$ survives.
Finite null $\Delta L = 2$ search	Bounded non-observation	Cannot prove exact all-operator preservation.

Candidate	What it actually selects	Constraint disposition
Secure positive $\Delta L = 2$ record	Failure of the exact additive distinction	Can certify $\text{Stand}_{\Delta L} = 0$ after target audit.
Certified carrier plus all-operator protector	Preservation of the exact additive distinction	Can reopen $\text{Stand}_{\Delta L} = 1$ only in the certified expanded target.
ΔL -standing verdict	Exact preservation or violation under the occupied mass role	Unique exact-character coordinate.

Theorem 12.1 (NMC-T7: exact-character selector uniqueness by exhaustion). *Let Q be a proposed standing-bearing selector for exact Dirac versus Majorana-compatible character. If Q preserves the neutrino mass-role target under gauge, field-carrier, boundary, quotient, low-energy, and envelope discipline, then Q is equivalent at the exact-character target to a certification of*

$$\text{Stand}_{\Delta L, \mathcal{E}}(\mu_D(\Pi_\nu)) \in \{0, 1\} \quad (39)$$

inside a selected envelope. Every other candidate in Table 7 acts on a different target, lacks standing, quotient-collapses, or changes scope.

Proof. Lemma 8.3 fixes the exact character axis independently of the selector list. Oscillation, scale, PMNS coordinates, diagonalization, and scalar mass values do not act on that axis. Availability and field admission do not certify occupancy or protection. Naturalness has no gate authority. Ultraviolet names quotient-collapse unless a declared witness survives. Finite null searches do not establish an exact zero. A secure violating record and an all-operator protector matter precisely because they certify opposite values of $\text{Stand}_{\Delta L}$. Exhaustion leaves the standing verdict as the unique exact-character coordinate. $\square$

12.3 General selector normal form

Theorem 12.2 (NMC-T8: neutrino-mass exact-character selector normal form). *In the broad target $\mathfrak{D}_\nu$, a global exact-character selection has the necessary normal form*

$$S_\nu = (\text{EnvSel}_\nu(\mathcal{E}), \text{Stand}_{\Delta L, \mathcal{E}}(\mu_D(\Pi_\nu))). \quad (40)$$

An envelope must be selected, or the same standing verdict must be certified invariant across all live envelopes.

Proof. Theorem 12.1 identifies the only exact-character coordinate. Different field and operator envelopes are not same-envelope rivals, so a global statement also requires a lawful envelope verdict. Criterion 2.5 then requires a certified occupied survivor consistent with the standing verdict and closure of every inequivalent same-target rival. $\square$

12.4 Low-energy Standard-Model-field envelope selection by impossibility

Proposition 12.3 (NMC-P10: corpus-certified low-energy Standard-Model-field envelope selection). *Within $\mathfrak{D}_{\nu,0}^{\text{LE}}$,*

$$W_{\text{osc}}^{m, \text{PMNS}} \wedge \text{SMFieldLoad} \wedge d \leq 5 \implies \text{EnvSel}_\nu^{\text{LE}}(\mathcal{E}_5) = 1. \quad (41)$$

Proof. By Theorem 4.5, the current same-target low-energy Standard-Model-field envelope candidates are $\mathcal{E}_0$ and $\mathcal{E}_5$. The mass-role target is certified by Theorem 4.2. The minimal renormalizable obstruction eliminates $\mathcal{E}_0$ as a complete realization once the mass-role witness is admitted. The declared $d \leq 5$ Standard-Model-field comparison class and Theorem 7.2 leave one available active-neutrino mass-role class in $\mathcal{E}_5$, and Corollary 7.5 certifies its occupancy with nontrivial quotient-spectrum support. The current corpus snapshot supplies no certified additional carrier/protector package. Given these facts together, the only remaining certified low-energy Standard-Model-field envelope in the bivalent envelope-status partition is $\mathcal{E}_5$. $\mathcal{E}_{\nu_R}$ and $\mathcal{E}_+$ are not silently excluded competitors: they add carrier or symmetry load and become certified only through the explicit admission routes fixed by $\mathfrak{D}_{\nu,+}^{\text{LE}}$ and Definition 2.7. $\square$

12.5 Exact-Dirac certified absence

Theorem 12.4 (NMC-T9: snapshot-relative exact-Dirac certified absence in $\mathfrak{D}_{\nu,0}^{\text{LE}}$). *In the current corpus-certified low-energy target,*

$$\text{NoPkg}_{\nu}^{\text{snapshot}}(\text{CarrierPkg}_{\nu_R}^{\text{cert}} \wedge \text{ProtectorPkg}_{\Delta L}^{\text{cert}}) \implies \text{Surv}_{\mathfrak{D}_{\nu,0}^{\text{LE}}}^{\text{cert}}(\text{ExactDirac}) = 0. \quad (42)$$

Under the joint acceptance criterion of Definition 2.7, the integrated current corpus snapshot supplies no certified Dirac-completing field carrier, occupied Dirac response, and standing-bearing protector covering every admitted $\Delta L = 2$ route inside $\mathfrak{D}_{\nu,0}^{\text{LE}}$.

Proof. Theorem 8.5 makes the carrier/protector package necessary for exact Dirac standing. $\mathfrak{D}_{\nu,0}^{\text{LE}}$ contains the currently certified Standard-Model light field load and no certified N_R or equivalent Dirac-completing carrier. A merely written or hidden carrier is not an operative field-role occupant by R6. A displayed zero, imposed label, or finite null search is not a persistence-bearing protector by the persistence-regime and all-operator closure results. A future source that certifies a carrier/protector package belongs to the update protocol $\mathfrak{D}_{\nu,+}^{\text{LE}}$ and triggers a new audit; it does not refute the present theorem relative to $\mathfrak{D}_{\nu,0}^{\text{LE}}$. Thus no exact-Dirac certified same-target survivor remains in the current corpus-snapshot target $\mathfrak{D}_{\nu,0}^{\text{LE}}$. $\square$

12.6 Certified low-energy exact-character collapse

Theorem 12.5 (NMC-T10: certified low-energy Majorana-compatible endpoint). *Within $\mathfrak{D}_{\nu,0}^{\text{LE}}$,*

$$\text{Sel}_{\mathfrak{D}_{\nu,0}^{\text{LE}}}(\text{MajoranaCompatible}_{\text{LE}}) = 1, \quad \text{Surv}_{\mathfrak{D}_{\nu,0}^{\text{LE}}}^{\text{cert}}(\text{ExactDirac}) = 0. \quad (43)$$

Selection is indexed to the evaluated target instance. The theorem selects $\text{MajoranaCompatible}_{\text{LE}}$ in $\mathfrak{D}_{\nu,0}^{\text{LE}}$; a future carrier/protector certificate would instantiate $\mathfrak{D}_{\nu,+}^{\text{LE}}$ and trigger a new theorem instance, not a contradiction of the present one.

Proof. Theorem 4.5 first fixes $\mathfrak{D}_{\nu,0}^{\text{LE}}$ from the admitted PMNS mass-driven witness family, currently certified Standard-Model carrier inventory, and low-energy witness quotient without selecting a character branch. The mass-role target is certified by Theorem 4.2, and Theorem 12.3 then selects $\mathcal{E}_5$ inside that fixed low-energy Standard-Model-field target. The dimension-five exhaustion supplies the unique Weinberg class, and Corollary 7.5 certifies its nontrivial occupied residue. That residue violates the exact additive persistence relation and is therefore Majorana-compatible by Lemma 8.3.

Theorem 12.4 eliminates exact Dirac as a certified same-target survivor of $\mathfrak{D}_{\nu,0}^{\text{LE}}$. Every unwitnessed richer-scope route is either low-energy equivalent to the occupied residue or an update handled by $\mathfrak{D}_{\nu,+}^{\text{LE}}$. Criterion 2.5 therefore leaves one certified occupied exact-character quotient in the current corpus-snapshot target. $\square$

Corollary 12.6 (NMC-C10: richer-scope exact-Dirac rescue requires a new certified target). *An exact-Dirac theory can contest the current source-state target instance of Theorem 12.5 only by certifying a Dirac-completing field carrier, an all-operator protector, and a surviving witness that triggers the update protocol $\mathfrak{D}_{\nu,+}^{\text{LE}}$ beyond $\mathfrak{D}_{\nu,0}^{\text{LE}}$.*

Corollary 12.7 (NMC-C11: Majorana-compatible does not select ultraviolet origin). *Theorem 12.5 selects the low-energy exact-character class, not a unique type-I, type-II, type-III, radiative, or other ultraviolet completion.*

Corollary 12.8 (NMC-C12: pure high-energy Majorana structure is not implied). *Mixed or richer-scope systems may project to the selected Majorana-compatible low-energy residue. Rank, scale, block profile, and pseudo-Dirac resolution remain separate targets.*

12.7 Deletion audit

A new source-state and target-instance audit is triggered under exactly three kinds of added certificate:

1. a certified Dirac-completing carrier and all-operator protector, which defines an expanded target;
2. a surviving richer-scope witness in $\mathcal{W}_{\text{low}}$, which prevents quotient collapse; or
3. a failure of the mass-driven or operator-basis premises, which changes the low-energy target itself.

Naturalness, coefficient stipulation, hidden carriers, and finite null searches do not reopen the branch.

13 Empirical witness-channel map

Empirical channels are possible certificate inputs or target-expansion witnesses; they are not definitions of the selector.

Table 8: Empirical interfaces to the certified low-energy endpoint.

Channel	Direct target	Effect on the theorem
Oscillation	Mass-driven spectrum and mixing support	Certifies the mass-role target; does not reverse exact character.
Direct beta decay	Absolute effective mass scale	Adds a scale anchor; does not certify $\text{Stand}_{\Delta L} = 1$.
Cosmological inference	Mass sum under transport assumptions	Constrains scale/model transport; not an exact-character certificate.

Channel	Direct target	Effect on the theorem
Secure positive $0\nu\beta\beta$ or other $\Delta L = 2$ record	Violation of the additive persistence relation	Strengthens the Majorana-compatible low-energy endpoint after mechanism/target audit; does not uniquely select UV origin.
Finite null $\Delta L = 2$ search	Bounded observation	non- Does not certify exact Dirac or an all-operator zero.
Heavy neutral lepton or partner-field detection	New field-carrier load	May certify a richer envelope and trigger reevaluation of $\mathfrak{D}_{\nu,0}^{\text{LE}}$.
Standing-bearing gauged or discrete protector evidence	All-operator exact protection	Could certify a new exact-Dirac target if the carrier and operator tower are jointly closed.
Correlated nonunitarity, flavor, collider, or cosmological residue	Surviving ultraviolet witness	Can prevent quotient collapse and discriminate origin.

Proposition 13.1 (NMC-P7: secure violation excludes exact Dirac). *A secure occupied $\Delta L = 2$ process contradicts preservation of Λ_ν and therefore excludes exact Dirac standing in the complete realized target. It does not by itself select a unique low-energy mechanism or ultraviolet origin.*

Corollary 13.2 (NMC-C13: positive neutrinoless double-beta decay is not a unique-origin selector). *Even a secure positive record establishes a violation class rather than a unique mediator. Origin selection still requires correlated witnesses surviving $\mathcal{W}_{\text{low}}$.*

Proposition 13.3 (NMC-P8: finite null searches do not certify exact Dirac). *A finite null search constrains the tested process and parameter region but cannot prove that every admitted $\Delta L = 2$ operator vanishes or is protectively forbidden.*

Future evidence can instantiate an updated source state or expanded target instance only by supplying the missing carrier–protector package or by introducing a richer-scope witness that survives the declared low-energy quotient. Naturalness, coefficient stipulation, and null non-observation do not perform that work.

14 Relation to the AASC source program

14.1 Imported dependency and role stack

The upstream-to-downstream dependency is explicit:

$$K_D \implies \text{NonNullRealizedPersistence} \implies [P_D]_{\equiv \text{Pers}} \implies \Phi_D \implies \Pi_D \implies \mu_D(\Pi) \implies \mu_D(\Pi_\nu). \quad (44)$$

R1 supplies the non-null realized-persistence floor; persistence-regime realization supplies consequence-retaining participation; R6 supplies field-carrier mediation; the intervening particle-class layer is imported through the realized-physics arc; R9 supplies the mass-role profile. The neutrino-specific manuscripts then supply the oscillation witness, matrix and quotient discipline, bridge/Yukawa status, operator envelopes, shape/scale firewalls, and charge/anomaly constraints [16, 17, 18, 19, 1, 5, 6, 7, 8, 3].

No arrow is reversed. A Dirac, Majorana, or Weinberg model may realize the mass role; it does not define what mass-role standing, target identity, or lawful selection mean.

14.2 Constraint results established here

The contribution of this paper is the following constraint package:

1. a positive neutrino mass-role target-fixation theorem;
2. a formal calculus separating realized occupancy, certified occupancy, survival, occupied survival, selection, equivalence, and richer scope;
3. fixed-envelope exhaustion of $\mathcal{E}_0$, $\mathcal{E}_5$, and $\mathcal{E}_{\nu_R}$;
4. an all-operator pure-Dirac protection theorem;
5. a declared-witness low-energy quotient rule for richer-scope origins;
6. exhaustion of proposed character selectors and uniqueness of the ΔL -standing coordinate;
7. the selector normal form $S_\nu = (\text{EnvSel}_\nu, \text{Stand}_{\Delta L})$ and its application inside $\mathfrak{D}_{\nu,0}^{\text{LE}}$; and
8. the corpus-certified low-energy Majorana-compatible exact-character endpoint.

The standard field-theory identities intentionally fix the arena. AASC contributes the role, promotion, selector, and endpoint theorems.

14.3 Compatibility with later numerical neutrino work

A later source may lawfully add ordering, ratio, shape, or absolute-scale results. Such results can refine the mass-spectrum rows of the present ledger. They do not automatically alter the character theorem. To do so, the later result must contain a source-certified predicate that distinguishes exact lepton-number preservation from violation or selects the operator envelope. Numerical mass values shared by Dirac and Majorana models do not supply that predicate.

15 Scope boundaries and theorem interpretation

15.1 AASC dependence as theorem class

AASC dependence fixes the theorem class rather than supplying an alternative result. A contrary same-target construction must invalidate the target-adequacy-to-kernel passage, a load-bearing dependency, the neutrino-specific specialization, or the final branch closure; alternatively, it must realize the faithful same-regime counterexample conditions. A model written in a different field envelope, a change of terminology, or an unrelated proof method leaves the same-target theorem unchanged. Theorem 1.2 and Section 16 give the structural alternatives.

15.2 Weinberg-operator sufficiency

It is the unique active-neutrino mass operator in the Standard-Model-field effective envelope through dimension five. That is a strong envelope-relative result. It is not a theorem that the realized comparison arena contains no singlets, no higher operators, and no protective structure. The paper preserves the full Weinberg uniqueness result while proving the impossibility of promoting it to a universal character verdict without an envelope selector.

15.3 Standard field-theory inputs and the theorem object

The operator and mass-matrix identities are comparison inputs rather than novelty claims. The theorem object is the standing of the inferences made from those facts. Standard identities do not by themselves distinguish availability from occupancy, a chosen zero from a protected zero, ultraviolet multiplicity from low-energy character multiplicity, or an envelope-relative singleton from a global singleton. The AASC status calculus and no-promotion theorem supply that missing constraint result.

15.4 Kernel dependency and proof direction

The kernel supplies the minimal necessity conditions for determinate same-domain construction. A comparison cannot assert that two models concern the same target, that a result is reusable, that one continuation is licensed, or that an exclusion has not been erased by repair unless reference, standing, admissibility, and irreversibility are already in force. The physical analysis is a specialization downstream of that target-adequacy result; reversing the dependency lets a candidate define the selector by which it wins.

15.5 Naturalness is not a standing selector

Naturalness, coefficient size, and economy are not standing selectors unless translated into a precise theorem. A gauge-singlet Majorana term is available without electroweak suppression, which makes pure Dirac structure unprotected in the generic singlet envelope. The constraint conclusion is not an aesthetic preference for Majorana mass. It is the impossibility of selecting exact Dirac as a certified same-target survivor of $\mathfrak{D}_{\nu,0}^{\text{LE}}$ without the required protector package.

15.6 Right-handed-neutrino availability is not exact-Dirac selection

Right-handed neutrinos make a Dirac Yukawa operator possible. Because they are gauge singlets, they also make singlet Majorana terms possible. The generic mass matrix is therefore mixed. A Dirac-only endpoint is a protected exact-zero branch, not the automatic consequence of adding N_R .

15.7 Positive neutrinoless double-beta decay and mechanism attribution

A secure positive result would establish lepton-number violation and exclude exact Dirac-only standing. It would not by itself prove that the amplitude is dominated by exchange of the observed light neutrinos or select a unique ultraviolet completion. The process target, effective operator, nuclear response, and matching map must be audited separately.

15.8 Null neutrinoless double-beta searches and exact-Dirac protection

A finite null search does not prove that every Majorana block is exactly zero. The pure-Dirac theorem demands a positive protection certificate, which is categorically stronger than non-observation in one channel.

15.9 Numerical mass values are a distinct target

Numerical mass reconstruction is a distinct theorem target. CNF6, NNR3, and NNR0.8 separate matrix support, quotient spectrum, difference space, shape, and absolute scale. The operator-character theorem is upstream of numerical readout. Adding unsupported numbers would violate rather than strengthen the constraint formalism.

15.10 Exhaustive classification as proof

Classification by exhaustion is the proof form. The paper goes beyond a residual survivor list: it fixes the mass role, exhausts false selectors, derives the selector normal form, and applies it inside $\mathfrak{D}_{\nu,0}^{\text{LE}}$. Bivalence, deletion of $\mathcal{E}_0$, uniqueness of the occupied Weinberg residue, and the certified absence of a Dirac carrier–protector package collapse the low-energy exact-character branch inside $\mathfrak{D}_{\nu,0}^{\text{LE}}$ to the Majorana-compatible endpoint. The scope boundary is part of the theorem statement and leaves the closure inside the declared target unchanged.

15.11 Global lepton number requires standing

It can be postulated as a model assumption. It becomes a selector only when it is certified as the standing-bearing protector of the exact neutrino/antineutrino distinction across every admitted $\Delta L = 2$ route, anomaly, quantum correction, matching step, and transport operation. The distinction is between constructing a Dirac model and certifying $\text{Stand}_{\Delta L} = 1$.

15.12 Corpus-certified absence follows the exact-Dirac burden theorem

The endpoint is not inferred from an unrecorded absence. The exact-Dirac burden is proved first, and the failure to discharge it is established by a reproducible corpus-wide carrier–protector audit. Spectral, scale, basis, availability, origin, preference, and sensitivity candidates fail to act on exact character. Only preservation or violation of the lepton-number-like persistence distinction changes the exact Dirac–Majorana standing verdict.

15.13 Lepton-number notation and the additive persistence relation

A conventional symbol L has no selector authority. The theorem uses a role, not a label: the quotient-stable additive distinction between neutrino and antineutrino continuation under the occupied mass response. If no admitted operation preserves such a distinction, the standing value is zero irrespective of notation. If an all-operator protector preserves it, the standing value is one.

15.14 Sterile-neutrino fits do not select an envelope

A fit parameter does not certify a standing-bearing carrier. A sterile route must close source production, propagation, mixing, detection, systematics, cosmological transport where used, and anomaly/gauge compatibility. Until then it is a richer-scope hypothesis, not an imported envelope-selection theorem.

16 Same-target closure, weakening resistance, and kernel-denial consequences

The following closure layer adapts the kernel-first same-target closure and weakening-resistance architecture developed for AASC endpoint proofs [15]. It does not add a new premise to the neutrino theorem. It records the invariants that a same-target counterexample must preserve and the structural losses incurred when a kernel role is denied.

16.1 Cost of kernel denial

Lemma 16.1 (NMC-L3: cost of kernel denial). *A denial of the kernel while retaining the same low-energy neutrino mass-character target must either concede the kernel work under different vocabulary or abandon at least one target-adequacy condition. Abandoning a condition incurs the corresponding loss recorded in Table 9.*

Table 9: Cost of kernel denial in $\mathfrak{D}_{\nu,0}^{\text{LE}}$.

Denied adequacy condition	Kernel role lost	Cost to the neutrino target
Target determinacy	Reference	No fixed object remains: mass character can drift among oscillation spectrum, mass scale, operator availability, ultraviolet origin, empirical channel, or notation.
Role evaluability	Standing	Dirac, Majorana-compatible, mixed, Weinberg, sterile, and protector claims lose branch standing and reduce to model names, inscriptions, or bookkeeping.
Act-time finality	Irreversibility	A failed exact-Dirac branch may be repaired later by coefficient stipulation, hidden-carrier addition, post-hoc protector import, or scope shift while being treated as the same evaluated act.
Same-regime delity	fi- Admissibility boundary	$\mathcal{E}_5$, $\mathcal{E}_{\nu_R}$, and $\mathcal{E}_+$ may be interchanged without a declared target change; carrier addition, protector addition, and richer-scope rescue become uncontrolled.

Proof. The four rows are the neutrino specialization of target determinacy, step or role evaluability, act-time finality, and same-regime fidelity. If none is denied, the proposed kernel weakening retains the governance work named by reference, standing, irreversibility, and admissibility. If one is denied, the corresponding same-target good is lost. A later device that restores the lost good either performs governance-equivalent work, introduces fresh authority, or changes the comparison domain. No cost-free third position remains. $\square$

16.2 The core low-energy adequacy packet and strict weakening

Definition 16.2 (Core low-energy neutrino adequacy packet). *The packet $\mathcal{C}_{0,\nu}^{\text{LE}}$ consists of the fixed target $\mathfrak{D}_{\nu,0}^{\text{LE}}$, the mass-role target $\mu_D(\Pi_\nu)$, the mass-driven witness $W_{\text{osc}}^{m,\text{PMNS}}$, the declared low-energy*

witness family $\mathcal{W}_{\text{low}}$, field-carrier and mass-role discipline, the exact-character selector $\text{Stand}_{\Delta L}$, act-time finality, and same-regime fidelity.

Definition 16.3 (Dirac-rescue-permissive weakening). *A strict weakening $\mathcal{W}_\nu^{\text{weak}}$ preserves $\mathcal{C}_{0,\nu}^{\text{LE}}$ if it claims to retain the same low-energy target, mass role, witness family, selector, and fixed-domain discipline while permitting exact Dirac preservation without a certified Dirac-completing field carrier and an all-admitted- $\Delta L = 2$ protector.*

Theorem 16.4 (NMC-T12: no strict same-target weakening permits exact-Dirac rescue). *No strict weakening $\mathcal{W}_\nu^{\text{weak}}$ preserves $\mathcal{C}_{0,\nu}^{\text{LE}}$ while certifying exact Dirac without $\text{CarrierPkg}_{\nu_R}^{\text{cert}} \wedge \text{ProtectorPkg}_{\Delta L}^{\text{cert}}$.*

Proof. Exhaust the possible weakening routes. A hidden or merely written N_R fails field-carrier certification. A chosen zero in a Majorana block is a parameter stipulation, not persistent protection. An imposed global label is skin unless it carries standing through every admitted operation. Naturalness is a ranking, not a selector. A finite null $\Delta L = 2$ search has bounded sensitivity and cannot certify the admitted operator tower. Finite-order checks in an unbounded effective envelope do not establish exact protection. A richer-scope carrier or symmetry changes the target unless a surviving witness and complete carrier/protector package are certified. A route that changes no branch-standing verdict is bookkeeping. The remaining route would preserve every component of $\mathcal{C}_{0,\nu}^{\text{LE}}$ while violating the exact-character burden theorem, contradicting Theorem 8.5. Hence the weakening either collapses extensionally to the original packet, imports inadmissible authority, or exits the target. $\square$

Table 10: Weakening-resistance audit for exact-Dirac rescue.

Proposed weakening	Missing or imported work	Disposition
Hidden or symbolic N_R	Operative standing	field-carrier Inert or syntax-only until carrier-certified
$M_R = m_L = 0$ by choice	Persistent all-operator protection	Parameter point, not exact protection
Uncertified global lepton number	Standing-bearing authority	Skin or richer-scope rule load
Naturalness or minimality	Exact-character authority	gate au- Ranking only; nonselector
Finite null $\Delta L = 2$ search	Tower-level exclusion	Bounded non-observation; nonselector
Finite-order EFT check	Unbounded protector theorem	Incomplete for exact character
Richer-scope carrier or symmetry	Same-target fidelity	New target unless witness-certified
No branch-status effect	Independent governance work	Bookkeeping

16.3 No one-step retyping

Lemma 16.5 (NMC-L4: no one-step retyping from oscillation to exact character). *The final Majorana-compatible endpoint in $\mathfrak{D}_{\nu,0}^{\text{LE}}$ is not inferred directly from oscillation, nonzero mass, Weinberg availability, or absence of a written singlet. The proof factors through the ordered chain*

$$\begin{aligned}
W_{\text{osc}}^{m,\text{PMNS}} &\implies \text{RoleTarget}_{\mathfrak{D}_{\nu,0}^{\text{LE}}}^{\text{cert}}(\mu_D(\Pi_\nu)), \\
\mathcal{E}_0 &\implies \text{no active-neutrino mass-role occupant}, \\
\mathcal{E}_5 &\implies \text{unique low-energy SM-field occupant}, \\
\text{ExactDirac} &\implies \text{CarrierPkg}_{\mathfrak{S}_{\nu_R}}^{\text{cert}} \wedge \text{ProtectorPkg}_{\Delta L}^{\text{cert}}, \\
\neg(\text{CarrierPkg}_{\mathfrak{S}_{\nu_R}}^{\text{cert}} \wedge \text{ProtectorPkg}_{\Delta L}^{\text{cert}}) &\implies \text{Surv}_{\mathfrak{D}_{\nu,0}^{\text{LE}}}^{\text{cert}}(\text{ExactDirac}) = 0, \\
\text{LowEnergyClosure} &\implies \text{Sel}_{\mathfrak{D}_{\nu,0}^{\text{LE}}}(\text{MajoranaCompatible}_{\text{LE}}) = 1.
\end{aligned} \tag{45}$$

Proof. Each arrow changes a different status type: target certification, envelope deletion, operator uniqueness, exact-Dirac burden, source-certified survival, and final selection. Omitting an intermediate step confuses a witness with an occupant, availability with occupancy, or source absence with universal nonexistence. The typed chain prevents those promotions. $\square$

16.4 Non-explosiveness of the low-energy collapse

Theorem 16.6 (NMC-T13: non-explosiveness of the low-energy exact-character collapse). *The conclusion*

$$\text{Surv}_{\mathfrak{D}_{\nu,0}^{\text{LE}}}^{\text{cert}}(\text{ExactDirac}) = 0$$

does not imply that no Dirac neutrino theory can be formulated or realized in any richer target. It excludes only exact Dirac as a certified same-target survivor of $\mathfrak{D}_{\nu,0}^{\text{LE}}$ without the required carrier/protector package.

Proof. Dirac seesaw models, radiative Dirac constructions, sterile sectors, gauged or discrete protectors, and other richer-scope proposals remain lawful model-building targets. They contest the low-energy endpoint only after their new carrier, protector, matching, and witness loads are certified and a new target is declared. Until then they are neither refuted globally nor admitted as same-target survivors. $\square$

16.5 Structural failure modes for same-target alternatives

Proposition 16.7 (NMC-P12: classification of same-target structural failure modes). *Any contrary same-target construction must invalidate at least one of the following load-bearing links:*

1. the target-adequacy-to-kernel passage;
2. the spectral-mass-role bridge;
3. the definition and fidelity of $\mathfrak{D}_{\nu,0}^{\text{LE}}$;
4. field-carrier certification;
5. the neutral-spinor exact-character normal form;

6. the exact-Dirac carrier/protector burden;
7. the reproducible carrier/protector audit;
8. dimension-five low-energy operator uniqueness;
9. the richer-scope witness quotient; or
10. the final exact-character collapse.

A construction outside these classes either changes the target or leaves the theorem chain intact unless it identifies a theorem-level failure in a cited bridge.

Proof. These links exhaust the proof spine from target adequacy to branch selection. Any same-target construction that preserves every link also preserves the conclusion. A construction that changes the field load, witness family, selector, or target is a scope extension rather than a counterexample to the same theorem. $\square$

16.6 Faithful same-regime counterexample conditions

Theorem 16.8 (NMC-T14: faithful same-regime counterexample conditions). *A faithful same-regime counterexample must preserve the fixed target $\mathfrak{D}_{\nu,0}^{\text{LE}}$, the mass-role target $\mu_D(\Pi_\nu)$, the witness family $\mathcal{W}_{\text{low}}$, the exact-character selector, act-time finality, and same-regime fidelity, while producing a certified exact-Dirac survivor or invalidating a named load-bearing theorem. An example that loses one of those preservation conditions, restores it through hidden authority, or changes the field envelope without certification is not a counterexample to the same result.*

Table 11: Faithful same-regime counterexample conditions.

Targeted link	Required same-target construction
Kernel denial	A target-adequate neutrino character evaluation preserving target, role evaluability, act-time finality, and redescription fidelity while omitting a named kernel role or governance-equivalent replacement.
Strict weakening	A weaker same-target regime permitting exact Dirac without a certified carrier/protector package and without losing any component of $\mathcal{C}_{0,\nu}^{\text{LE}}$.
Mass-role bridge	A PMNS mass-driven oscillation witness that changes propagation response but does not instantiate the transport-facing mass-role profile.
Exact-character selector	A neutral-spinor exact-character coordinate not reducible to additive particle/antiparticle persistence standing.
Exact-Dirac rescue	A certified N_R or equivalent carrier plus a standing-bearing protector covering every admitted $\Delta L = 2$ route.
Corpus audit	A source object meeting the joint carrier/protector acceptance criterion that the audit omitted or misclassified.
Richer-scope route	A route with a surviving $\mathcal{W}_{\text{low}}$ witness that remains the same target and avoids quotient collapse.

Targeted link	Required same-target construction
Endpoint alternative	A certified exact-Dirac survivor inside $\mathfrak{D}_{\nu,0}^{\text{LE}}$.

Proof. A contrary construction to a necessity or impossibility theorem must either break a premise-to-conclusion link or exhibit the target phenomenon without the claimed necessary structure. The preservation clauses prevent target drift from masquerading as counterexample. Any restoring selector, carrier, tolerance, or repair channel must itself be audited as governance work. $\square$

16.7 Exhaustion of same-target alternatives

Theorem 16.9 (NMC-T15: exhaustion of remaining same-target alternatives). *After the target lock, mass-role bridge, exact-character normal form, carrier/protector burden, corpus audit, and low-energy quotient rule are in force, every same-target alternative must establish at least one of the following:*

1. $\mathfrak{D}_{\nu,0}^{\text{LE}}$ is not the declared current low-energy target or excludes a certified same-target carrier;
2. $W_{\text{osc}}^{m,\text{PMNS}}$ does not certify $\mu_D(\Pi_\nu)$;
3. exact neutral-spinor character is not governed by $\text{Stand}_{\Delta L}$;
4. exact Dirac does not require $\text{CarrierPkg}_{\nu_R}^{\text{cert}} \wedge \text{ProtectorPkg}_{\Delta L}^{\text{cert}}$;
5. the required package exists in the integrated corpus;
6. a richer-scope route remains same-target and avoids the $\mathcal{W}_{\text{low}}$ quotient; or
7. exact Dirac survives in $\mathfrak{D}_{\nu,0}^{\text{LE}}$ despite the preceding exclusions.

A change of terminology or a demand for a different proof class does not instantiate any branch of this same-target exhaustion.

Proof. The seven alternatives correspond respectively to target failure, bridge failure, selector failure, burden failure, audit failure, quotient failure, and final-selection failure. They are exhaustive because every inference in the proof spine belongs to one of those classes. If none occurs, the endpoint theorem is preserved. $\square$

17 Conclusion

AASC has been applied as a constraint formalism. The kernel fixes the minimum conditions for meaningful same-target construction; realized-persistence, field-carrier, and mass-role theorems prevent the neutrino target from being replaced by trace, notation, hidden carrier, primitive scalar, or model preference. In the standard coherent PMNS mass-driven scope, oscillation fixes the neutrino mass-role target but does not itself identify a field-theoretic occupant.

The operator and carrier audit closes the declared low-energy exact-character target. The minimal renormalizable Standard-Model envelope has no active-neutrino mass-role occupant. The Standard-Model-field envelope through dimension five has the unique Weinberg residue. Exact Dirac preservation requires a certified Dirac-completing field carrier and a standing-bearing protector

excluding every admitted $\Delta L = 2$ route. The singlet remarks in the corpus establish availability and alternative model routes; they do not certify an occupied carrier or protector. The reproducible audit finds no joint carrier–protector package satisfying the acceptance criterion.

The endpoint is therefore not a no-singleton boundary and not merely an identified selector coordinate. Relative to $\mathfrak{D}_{\nu,0}^{\text{LE}}$, exact Dirac is excluded from the certified same-target survivor set, unwitnessed richer-scope routes quotient-collapse, and the selected low-energy exact-character endpoint is

$$\text{Sel}_{\mathfrak{D}_{\nu,0}^{\text{LE}}}(\text{MajoranaCompatible}_{\text{LE}}) = 1. \quad (46)$$

This is a Majorana-compatible low-energy result. It does not select a unique ultraviolet mechanism, a pure-Majorana ultraviolet block structure, or numerical neutrino masses. An exact-Dirac rescue must enter as a newly certified target with its own carrier, occupied Dirac response, protector, operator-tower closure, matching, and surviving witness package.

The theorem is AASC-internal by construction, and that dependency is explicit rather than concessive. A contrary same-target construction must derive a kernel role from below, break a load-bearing link in the kernel-to-target chain, identify a source-audit failure, or satisfy the faithful same-regime counterexample conditions of Theorem 16.8. Without one of those results, the cost-of-denial and anti-weakening theorems leave the low-energy endpoint unchanged.

A Complete operator-exhaustion ledger

Table 12: Operator classes, standing conditions, and endpoint status.

Class / lowest envelope	Fields or schematic operator	Lepton-number and character status	Occupancy status	Principal blocker or selector
$\mathcal{C}_0; \mathcal{E}_0$	No active-neutrino mass operator	Exact but trivial for the mass target; massless	Contradicted as a complete account by the admitted nontrivial oscillation witness	Oscillation spectral witness
$\mathcal{C}_D; \mathcal{E}_{\nu_R}$	$\bar{L} \tilde{H} N_R$	Dirac only if every Majorana term is forbidden	Available; nonzero coefficient required for mass	Protective standing and Yukawa occupancy
$\mathcal{C}_W; \mathcal{E}_5$	$(L \cdot H)^T C (L \cdot H) / \Lambda$	Conventional $\Delta L = 2$; active Majorana	Conditionally occupied with nontrivial quotient-spectrum support if $\mathcal{E}_5$ is complete	Envelope completeness; coefficient and scale remain
$\mathcal{C}_R; \mathcal{E}_{\nu_R}$	$N_R^T C N_R$	$\Delta L = 2$; heavy/light Majorana contribution	Available; coefficient unselected	Protector can forbid; a secure $\Delta L = 2$ record can support
$\mathcal{C}_{\text{mix}}; \mathcal{E}_{\nu_R}$ or $\mathcal{E}_+$	m_D with M_R and/or m_L	Broken lepton number; mixed Majorana eigenstates	Generic when allowed coefficients are nonzero	Texture, rank, and scale witnesses
$\mathcal{C}_{\text{pD}}; \mathcal{E}_{\nu_R}$ or $\mathcal{E}_+$	Near-paired Majorana spectrum relative to readout	Approximately preserved response; pseudo-Dirac pairs	Readout-relative regime, not a fundamental bilinear class	Quotient-stable pair splitting and declared resolution

Class / lowest envelope	Fields or schematic operator	Lepton-number and character status	Occupancy status	Principal blocker or selector
$\mathcal{C}_I; \mathcal{E}_{\nu_R}/\mathcal{E}_+$	Heavy singlet fermions	Usually Majorana low-energy residue	Model-conditional	Heavy-state or correlated-operator witness
$\mathcal{C}_{II}; \mathcal{E}_+$	Scalar triplet	Majorana	Model-conditional	Triplet representation and decay witness
$\mathcal{C}_{III}; \mathcal{E}_+$	Fermion triplet	Majorana	Model-conditional	Triplet production and gauge witness
$\mathcal{C}_{\text{rad}}; \mathcal{E}_+$	Additional loop carriers	Dirac or Majorana, model-dependent	Model-conditional	New fields, symmetry, and loop-linked operators
$\mathcal{C}_{d>5}; \mathcal{E}_+$	Higher-dimensional operators	Dirac or Majorana, model-dependent	Model-conditional	Declared dimension and field envelope

B Gauge and representation checks

B.1 Minimal-field obstruction

The active lepton doublet transforms as $L \sim (\mathbf{1}, \mathbf{2}, -\frac{1}{2})$ and the Higgs doublet as $H \sim (\mathbf{1}, \mathbf{2}, +\frac{1}{2})$ in the conventional hypercharge normalization. A bare left-Majorana bilinear $L^T C L$ is not an electroweak singlet with the correct hypercharge. A renormalizable Dirac term requires a right-handed neutral singlet absent from $\mathcal{E}_0$. Therefore no $d \leq 4$ active mass term exists in that field load.

B.2 Dimension-five singlet

The combination of two lepton doublets and two Higgs doublets can be contracted to an electroweak singlet. Up to flavor coefficients, integration by parts, field redefinition, and index-contraction identities, it produces the unique Standard-Model effective operator at dimension five [24, 25]. After the neutral Higgs component acquires its bridge response, the operator yields a complex symmetric active-neutrino mass matrix.

B.3 Right-handed neutral singlet

A field $N_R \sim (\mathbf{1}, \mathbf{1}, 0)$ permits the Yukawa singlet $\bar{L} \tilde{H} N_R$. Its vanishing hypercharge also permits $N_R^T C N_R$. The charge-lattice source identifies a $Y = 0$ singlet as anomaly-inert, so adding it does not repair or disturb the charged hypercharge collapse by anomaly cancellation alone [10]. This is precisely why gauge and anomaly conditions do not protect the Dirac branch.

C Mass-matrix normal forms and invariants

C.1 Exact Dirac form

With $m_L = M_R = 0$,

$$\mathcal{L}_m = -\bar{\nu}_L m_D N_R + \text{h.c.} \quad (47)$$

is invariant under a common continuous phase rotation of ν_L and N_R . The singular values of m_D are basis invariant. Texture entries and left/right diagonalizing matrices are not individually standing until compared with the charged-current ledger.

C.2 Exact Majorana form

For a symmetric matrix M acting on left-chiral neutral fields,

$$\mathcal{L}_m = -\frac{1}{2}n_L^T C M n_L + \text{h.c.}, \quad M = M^T. \quad (48)$$

Takagi singular values are nonnegative invariants. The conventional lepton-number charge is broken whenever the corresponding Majorana block is nonzero.

C.3 Seesaw block reduction

When $\|M_R\| \gg \|m_D\|$ and M_R is invertible, block diagonalization yields the leading light matrix

$$m_\nu^{\text{eff}} = -m_D M_R^{-1} m_D^T + \mathcal{O}(m_D^4 M_R^{-3}). \quad (49)$$

This formula is conditional on the hierarchy and invertibility assumptions. It does not by itself establish the existence, scale, or occupancy of M_R . In the low-energy effective action, the result lies in the Weinberg class.

C.4 Rank bounds

For the leading type-I expression,

$$\text{rank}(m_\nu^{\text{eff}}) \leq \min\{\text{rank}(m_D), \text{rank}(M_R^{-1})\} \leq n_R. \quad (50)$$

The bound is useful for route-specific exclusion but cannot determine n_R without a complete spectral-rank witness and a restriction to the type-I route.

C.5 Pseudo-Dirac diagnostic

A basis-independent pseudo-Dirac diagnostic must be stated in terms of near-paired Takagi values, approximately off-diagonal charge structure, and small charge-violating invariants. Small entries in a preferred basis are insufficient. The resolution scale and propagation baseline are part of the diagnostic because a pair can be operationally Dirac-like over one regime and resolved as two Majorana states over another.

D Ultraviolet route audit

Table 13: Representative, non-exhaustive richer-scope route audit.

Route	New load	Matched light residue	Potential distinguishing witness	Forbidden over-read
Type I	Singlet fermions and heavy mass matrix	Weinberg operator at leading order	Heavy-state production, nonunitarity, flavor violation	Matching does not prove heavy singlets exist.
Type II	Scalar triplet and scalar couplings	Weinberg operator	Doubly charged scalar, triplet decays, scalar-potential residues	Majorana character does not identify a scalar triplet.
Type III	Fermion triplet	Weinberg operator	Gauge production and triplet decay pattern	Majorana character does not identify a fermion triplet.
Radiative Majorana	Extra fields, symmetry, and loop topology	Weinberg or higher operator	Correlated dark, flavor, or scalar operators	Loop suppression is not an explanation without carrier certification.
Inverse seesaw	Additional singlets and small $\Delta L = 2$ parameter	Mixed or pseudo-Dirac light and heavy sectors	Accessible heavy states, nonunitarity, small-splitting pattern	A small parameter is not protected unless the symmetry ledger closes.
Dirac seesaw or radiative route	Partner fields and exact protector	Effective Dirac Yukawa	New mediator and protector-linked signatures	Dirac character does not identify the mediator.
Extra-dimensional or nonlocal route	New geometry or non-local roles	Model-dependent	Kaluza–Klein tower, modified propagation, other residues	Richer scope cannot be counted as a same-domain repair.

E Source theorem dependency map

Table 14: Load-bearing source imports.

Source	Imported statement	Used in	Status	Boundary
Kernel source lock	Target adequacy forces kernel roles; mutual closure and same-target invariance	Proposition 2.1; Criterion 2.5	Canonical lineage, exact-object records, and provenance-marked proof extract	Complete proof used here is bundled in the extract and full foundational source
Foundational burden source	Mathematical-status lock, kernel derivation, weakening resistance, and same-target closure criteria	Propositions 1.1, 16.7; Theorems 1.2, 16.4–16.9	Method and verification dependency	The SAT endpoint conclusion is not imported
R1	Realized persistence cannot be reconstruction-null	Target non-nullity in Theorem 4.2	Proof dependency	No field, particle, or mass derivation imported
PERSIST	Irreversible standing-bearing continuation induces persistence-regime realization	No trace-only carrier or protector	Proof dependency	No concrete representative selected
R6	Distributed field-carrier mediation	Carrier-package burden and hidden-carrier deletion	Load-bearing	No Standard-Model species selected by the role theorem
R9	Mass/inertia as deformation-response profile $\mu_D(\Pi)$	Spectral-mass-role bridge	Load-bearing	No numerical mass or character imported
CS6C; NNR1	Mass-driven boundary-kernel and splitting witness	Lemma 3.2; Theorem 4.2	Proof dependency	No absolute offset or character selection
NNR3; NNR0.8	Shape/offset and anchor firewalls	Scale and anchor propositions	Proof dependency	No numerical shape or selector by anchor
CNF6; MGN5	CNF7; Matrix, spectrum, mixing, and basis quotients	Neutral-spinor and block-profile discipline	Proof dependency	No mass values from diagonalization
MGN3; MGN4	Bridge and Yukawa availability boundaries	Dirac slot and scale status	Proof dependency	Slot availability is not carrier certification
Charge; CS3; SM; Gen.	Field, anomaly, finite-alternative, and residual-load discipline	Envelope and protector audit	Proof dependency	Singlet remarks do not certify occupancy or protection
Weinberg; EFT	Unique SM-field dimension-five active mass class	Theorem 7.2	Standard input	Does not select a unique UV origin
Corpus carrier and protector audit	Reproducible search over all active source fields	Theorem 12.4	Load-bearing source audit	Source-relative to the integrated active corpus

F Claim and nonclaim register

Table 15: Principal claim register.

ID	Claim	Status	Primary dependency
NMC-P0	Kernel and role adequacy are upstream of neutrino-model comparison	Proved dependency lock	Kernel lineage, R1, PERSIST, R6, R9
NMC-P11	AASC dependence is a declared proof setting; contrary constructions must preserve the same target	Proved mathematical-status lock	Foundational burden source
NMC-T11	Treating the kernel as optional while retaining the same target is dependency inversion	Proved necessity theorem	NMC-P0, NMC-P11
NMC-K1	Selection is uniqueness over certified occupied survivor quotients	Governing criterion	Fixed-target calculus
NMC-L1	Mass-driven oscillation does not select exact character	Proved	CS6C, NNR1
NMC-L2	Neutral-spinor exact character is preservation/failure of an additive persistence relation	Proved normal form	Standard neutral-spinor bilinears, quotient discipline
NMC-T1	PMNS mass-driven oscillation certifies the neutrino mass-role target	Proved conditional on declared kernel	R1, PERSIST, R6, R9, CS6C
NMC-T16	Witness family, certified carrier inventory, and low-energy quotient fix $\mathfrak{D}_{\nu,0}^{\text{LE}}$	Proved target-fixation theorem	NMC-T1, R6, SM inventory, $\mathcal{W}_{\text{low}}$
NMC-T2	No active-neutrino mass operator exists in $\mathcal{E}_0$	Proved impossibility	SM envelope
NMC-T3	The Weinberg class is unique through dimension five	Envelope-relative theorem	EFT basis convention
NMC-T4	Availability does not imply certified occupancy	Proved status theorem	AASC calculus
NMC-T5	Exact Dirac requires a certified carrier-protector package	Burden theorem	R6, singlet/operator exhaustion
NMC-T6	Identical declared low-energy witness projections define one class	Quotient theorem	$\mathcal{W}_{\text{low}}$
NMC-T7	ΔL standing is the unique exact-character coordinate	Selector-exhaustion theorem	NMC-L2, NMC-T1–T6
NMC-T8	Global exact-character selection has the selector normal form ($\text{EnvSel}_{\nu}, \text{Stand}_{\Delta L}$)	Exhaustion theorem	Target and carrier discipline
NMC-P10	The low-energy Standard-Model-field target selects $\mathcal{E}_5$ by deletion and uniqueness	Envelope-selection proposition	NMC-T16, NMC-T1–T4
NMC-T9	No certified carrier-protector package in the current snapshot implies no exact-Dirac survivor in $\mathfrak{D}_{\nu,0}^{\text{LE}}$	Corpus-certified absence theorem	NMC-T5, corpus audit
NMC-T10	The selected low-energy exact-character endpoint in $\mathfrak{D}_{\nu,0}^{\text{LE}}$ is Majorana-compatible	Certified collapse theorem	NMC-T1–T9, NMC-P10
NMC-L3	Kernel denial either concedes equivalent work or pays a target-adequacy cost	Proved cost lemma	NMC-P0, NMC-T11
NMC-T12	No strict same-target weakening permits exact-Dirac rescue	Proved anti-weakening theorem	NMC-T5, NMC-T9
NMC-L4	The endpoint follows through a typed chain rather than one-step retyping	Proved route lemma	NMC-T1–T10
NMC-T13	The low-energy collapse does not prohibit richer-target Dirac theories	Proved non-explosiveness theorem	NMC-T9, NMC-T10

ID	Claim	Status	Primary dependency
NMC-P12	Contrary same-target constructions must invalidate a named load-bearing link	Proved failure-mode proposition	NMC-P11, NMC-T11
NMC-T14	A faithful counterexample preserves the same target while omitting required structure	Proved counterexample-conditions theorem	NMC-L3, NMC-T12
NMC-T15	Remaining same-target alternatives reduce to seven named failures	Proved exhaustion theorem	NMC-T11–T14
NMC-P7	Secure $\Delta L = 2$ excludes exact Dirac standing	Conditional interface proposition	Target map
NMC-P8	Finite null $\Delta L = 2$ does not prove exact Dirac	Proved nonselection	NMC-T5
NMC-N1	Absolute masses are derived	Explicit nonclaim	Shape/anchor firewall
NMC-N2	Every richer-scope model is enumerated	Explicit nonclaim	Open $\mathcal{E}_+$
NMC-N3	A unique ultraviolet route is selected	Explicit nonclaim	Low-energy quotient
NMC-N4	Leptogenesis is derived	Explicit nonclaim	Outside target

G Forbidden-promotion and endpoint register

Table 16: Forbidden promotions and required certificates.

Input	Forbidden promotion	Required certificate or endpoint
$W_{\text{osc}}^{m,\text{PMNS}}$	Specific operator or character occupant	Role-target certification only; envelope and occupant theorem still required
Unique dimension-five operator	Universal Majorana verdict over every extension	Low-energy target lock and $\text{EnvSel}_{\nu}^{\text{LE}}(\mathcal{E}_5) = 1$
Written N_R	Operative Dirac carrier	$\text{CarrierPkg}_{\nu_R}^{\text{cert}} = 1$
Displayed Majorana zero	Protected exact zero	$\text{ProtectorPkg}_{\Delta L}^{\text{cert}} = 1$ covering the declared operator tower
Finite null $0\nu\beta\beta$	Exact Dirac proof	All-operator protection theorem
UV model name	Distinct low-energy exact character	Surviving witness in $\mathcal{W}_{\text{low}}$
No carrier/protector package in $\mathfrak{D}_{\nu,0}^{\text{LE}}$	Ontological impossibility in every extension	Certified same-target absence only in the current snapshot; richer-scope rescue is handled by $\mathfrak{D}_{\nu,+}^{\text{LE}}$
Occupied Weinberg residue plus exact-Dirac deletion	Pure UV Majorana origin	$\text{MajoranaCompatible}_{\text{LE}}$ low-energy endpoint only
AASC-independence claim	Framework-neutral exemption from the proof	Lower derivation, named dependency break, or faithful same-regime counterexample

Input	Forbidden promotion	Required certificate or endpoint
Strict same-target weakening	Exact-Dirac rescue without carrier/protector burden	Theorem 16.4; route must exit the target or supply the missing certificates

H Notation and symbol index

Table 17: Principal notation.

Symbol	Meaning
K_D	AASC kernel ($\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D$)
$\mathfrak{D}_\nu^{\text{LE}}$	Low-energy active-neutrino exact-character target schema; $\mathfrak{D}_{\nu,0}^{\text{LE}}$ is the current corpus-snapshot instance and $\mathfrak{D}_{\nu,+}^{\text{LE}}$ is the admissible update protocol
$\text{TargetFix}(\mathfrak{D}_{\nu,0}^{\text{LE}})$	Witness-family, certified-carrier, and low-energy-quotient fixation of the current snapshot target
$\mu_D(\Pi_\nu)$	Neutrino mass-role deformation/transport-response profile
$W_{\text{osc}}^{m,\text{PMNS}}$	Standard coherent PMNS mass-driven oscillation witness
$\text{RoleTarget}_X^{\text{cert}}(R)$	The source/witness stack certifies role R as the target to be occupied
$\text{CarrierPkg}_{\nu R}^{\text{cert}}$	Certified Dirac-completing field-carrier and occupied Dirac response package
$\text{ProtectorPkg}_{\Delta L}^{\text{cert}}$	Certified all-operator $\Delta L = 2$ protection package
Λ_ν	Quotient-stable additive neutrino/antineutrino persistence relation
$\text{Stand}_{\Delta L, \mathcal{E}}(\mu_D(\Pi_\nu))$	Preservation or violation of Λ_ν under the occupied mass role
ExactDirac	Exact $\text{Stand}_{\Delta L} = 1$ branch
$\text{MajoranaCompatible}_{\text{LE}}$	Low-energy exact-character quotient in which the occupied active-neutrino residue fails exact preservation of Λ_ν ; not a unique ultra-violet representative
$\text{Occ}_X^{\text{real}}(C)$	Candidate class actually occupies a realized role
$\text{Occ}_X^{\text{cert}}(C)$	The theorem/witness stack certifies class occupancy
$\text{Surv}_X^{\text{cert}}(C)$	Candidate remains a source-certified survivor of target X
$\text{Sel}_X(C)$	Unique certified occupied survivor quotient
$\mathcal{W}_{\text{low}}$	Complete declared low-energy witness family
$\mathcal{C}_{0,\nu}^{\text{LE}}$	Core low-energy target, role, witness, selector, and fixed-domain adequacy packet
$\mathcal{W}_\nu^{\text{weak}}$	Alleged strict same-target weakening permitting exact-Dirac rescue
$\text{SameRegCE}_\nu^{\text{LE}}$	Faithful same-regime counterexample predicate for the low-energy target
$\mathcal{E}_0, \mathcal{E}_5, \mathcal{E}_{\nu R}, \mathcal{E}_+$	Minimal, dimension-five, singlet, and open richer-scope envelope classes

Symbol	Meaning
$L, H, \tilde{H}, N_R$	Lepton doublet, bridge doublet, conjugate bridge, and right-handed neutral singlet
$Y_\nu, \kappa/\Lambda, m_D, m_L, M_R$	Dirac Yukawa, Weinberg coefficient/scale, and neutral mass blocks
$\mathfrak{P}_D$	Standing-bearing all-operator protector
π_{low}	Projection or matching map to the low-energy target

I Reproducibility and audit protocol

The project archive accompanying this paper contains:

1. the canonical L^AT_EX source and compiled PDF;
2. the focused source PDFs and source texts used for proof-sensitive reuse;
3. a SHA-256 source inventory;
4. machine-readable dependency, claim-status, operator-exhaustion, empirical-selector, richer-scope, protective-standing, promotion-calculus, forbidden-promotion, consistency, and symbol ledgers;
5. filtered query-corpus snapshots for the source lineages used here;
6. build and PDF quality-assurance reports; and
7. a manifest recording file identities and SHA-256 hashes.

A theorem is marked “source imported” only when its supporting source is present in the project archive. The canonical standalone kernel lineage is represented here by exact accepted-object and version/dependency records; the archive does not represent the standalone kernel manuscript as a separately bundled complete source. The complete self-contained proof used for inspection in this article is bundled both in its full foundational-burden source and as a provenance-marked, page-exact kernel proof extract. External field-theory references support conventional operator and mechanism statements; they do not replace AASC source dependencies.

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